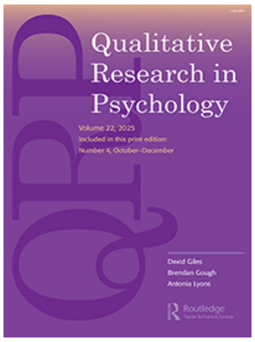

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The approach and application of analysing inductive and deductive datasets: a worked example using reflexive thematic analysis

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ABSTRACT

Braun and Clarke's reflexive thematic analysis (reflexive TA) has gained wide attraction since its conception in 2006. Reflexive TA is methodologically flexible with researchers making decisions, which support their philosophical positionings. Qualitative research publications are often criticised for their lack of detail on the analytical processes undertaken to develop the analysis. This level of detail is important for novice researchers in learning 'how to do' analysis. This paper provides a worked example of the analytical approach using reflexive TA when exploring parent treatment decision-making in poor-prognosis childhood cancer. Qualitative analysis was undertaken of interview transcripts at different decision timepoints incorporating inductive and deductive approaches resulting in within and across case analysis. Discussion reflecting on the challenges, iterative processes and contextual decisions made on and towards the data is provided. The aim is to share an example of 'how to' approach reflexive TA to support novice researchers in their learning.

KEYWORDS
Data analysis; methods; qualitative research; reflexive thematic analysis; induction; deduction; decision-making

Introduction

Thematic analysis (TA) is a method widely used in the analysis of qualitative research, yet it was considered 'poorly demarcated and rarely acknowledged' (Braun and Clarke 2006, 77) with inconsistencies in how TA is approached and conducted. There are different approaches to TA across the various methodological paradigms, with variations in how each of these paradigms is described (Braun and Clarke 2022b). Extensive reading is required by the researcher to be clear in how they are engaging with the various TA methods relevant to their methodological paradigm (Braun and Clarke 2022b). These paradigms include value and belief systems which contain ontological,

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epistemological and methodological assumptions (Braun and Clarke 2022b) which the researcher engages with. The methodological paradigm informs how the researcher approaches the whole research study from devising research questions, data collection and analysis through to writing up the findings. Thematic Analysis was initially defined as a 'method for identifying, analysing and reporting patterns (themes) within the data' (Braun and Clarke 2006, 79). However, Braun and Clarke reconceptualised their definition of TA to acknowledge that their approach now known as *reflexive* TA, is not a single method, and there are various ways to approach and do reflexive TA (Braun and Clarke 2022b, 9). For the researchers, this means they need to be clear in their methodological positioning and the purpose of their analysis to engage with the variations of reflexive TA.

Thematic analysis was first considered as a method of analysing qualitative data by Braun and Clarke in 2006. Braun and Clarke's initial conceptualisation of thematic analysis method detailed six iterative phases with methodological and theoretical flexibilities (Braun and Clarke 2006) allowing researchers to acknowledge the assumptions which inform their analysis. This approach to TA gained wide interest following the publication of their conceptual paper in 2006 which now has over 200,000 citations (Braun and Clarke 2006). However, Braun and Clarke's own reflection of their approach to TA saw misconceptions on how this was being interpreted with phases seen as a procedure that had to be followed precisely with little room for researchers' creativity and theoretical engagement (Braun and Clarke 2019a). This led to their revised approach, *reflexive* TA (Reflexive TA) (Braun and Clarke 2019b) which explicitly acknowledged the researchers' reflexivity and subjectivity in the analytical TA process and the importance of their methodological positioning which informs how they engage with data (Braun and Clarke 2024). Reflexivity requires the researcher to critically reflect on the personal and theoretical assumptions and discipline which shape their research design choices and knowledge produced (Clarke 2021; Braun and Clarke 2022b). Through reflexivity, a researcher questions who they are (Johns 2017) reflecting on themselves (Sparkes and Smith 2014) to consider what is happening within the data in relation to their assumptions, positionality, and how these influence their analytical developments. This active process contributes to the analysis and knowledge produced drawing on the subjectivity of the researcher. Subjectivity refers to the inherent qualities of a researcher, their values, identities, skills and experience. These qualities are seen as a resource in qualitative research as opposed to more quantitative approaches where they are considered as a bias (Braun and Clarke 2022b). The relationship a researcher has with the research subject or population is their positioning within the context of the research (Trainor and Bundon 2020). This positioning is also important to acknowledge as it influences how the data is produced and constructed. Reflexive TA acknowledges the importance of researcher subjectivity and reflexivity as resources for creativity and interpretation within the data (Braun and Clarke 2019a). This engagement with the data actively generates meaning-making in telling the story of the data (Braun and Clarke 2019a). This means that different researchers working with the same data may develop different themes and narratives. However, in doing so, the data should resonate with the reader who has knowledge and understanding on the phenomenon being researched.

To support researchers' approaches to reflexive TA, Braun and Clarke have made available a wealth of publications addressing important concepts such as quality and best practice in reflexive TA (Braun and Clarke 2020a, 2024), reviewing good and bad practices of reflexive TA (Braun and Clarke 2022a), can and should researchers use TA (Braun and Clarke 2020b) and conceptual and design thinking for TA (Braun and Clarke 2021). What all these publications offer is opportunities to engage with concepts of reflexive TA to develop a robust systematic approach to data analysis that allows for creativity, subjectivity and reflexivity with researcher engagement and immersion with the data. Braun and Clarke also hold webinars organised by SAGE publications for researchers sharing insights and interpretations of the reflexive TA method. Webinars such as 'Building good practice in reporting thematic analysis' https://www.youtube.com/live/En1HA_xwM4M?feature=share. (Braun and Clarke 2022a) and Common challenges in Thematic Analysis and how to avoid them with Virginia Braun & Victoria Clarke – YouTube (Braun and Clarke 2023) provide accessible engagement with Braun and Clarke answering questions and providing advice to researchers. Clarke has a YouTube channel with various reflexive TA lectures (Clarke 2017) Victoria Clarke – YouTube which is a great resource for novice researchers or those new to qualitative research. This accessibility to Braun and Clarke's work is noteworthy and should be shared widely as it provides opportunities for the qualitative research community to share their experiences and offer advice on the methods and approaches to reflexive TA. Additional publications by novice researchers, sharing their 'how to' approach to reflexive TA are essential contributions to this learning landscape. Together with Braun and Clarke's generosity, sharing 'how to' worked examples can help novice scholars with their own thinking, reflexivity, and analytical developments.

Two examples which have detailed their analytical approaches and application of reflexive TA have previously been published (Byrne 2021; Trainor and Bundon 2020). Both examples provide different ways to approach reflexive TA and reinforce the importance of acknowledging subjectivity and reflexivity in the analytical process. However, neither of these publications provide an example of the deductive orientation to analysis which is particularly important when analysing multiple datasets where researchers could be combining both inductive and deductive orientations. This paper adds to these examples with an analysis of multiple datasets over time combining inductive and deductive approaches. An inductive orientation captures meaning directly from what participants have said driven by the data, giving voice to participants' stories of their experiences (Braun and Clarke 2022b). In contrast, a deductive orientation is researcher or theory driven (Braun and Clarke 2022b) with interpretation of the data informed by theoretical frameworks, models or research literature. Combining these analytical orientations requires researchers to think carefully in how they engage in meaning-making of their data to ensure analytical depth and nuanced analysis. Expanding this learning landscape is essential. The availability of multiple examples provides variation of analytical approaches to reflexive TA giving novice researchers the confidence to see the iterative nature of this method which is informed by their subjective and reflexive practices. The uncertainty of starting out in reflexive TA and having the confidence to do this well is emphasised by Braun and Clarke (2022b) in their most recent textbook which acts as a tool to guide researchers through the process to become reflexive analysts. This is particularly important for novice researchers who are new to qualitative research as the terminology, concepts and approaches vary depending on the analytical method used. For example, grounded theory incorporates data saturation with data collection ceasing when data is producing no new theoretical insights (Glaser and Strauss 2008). In reflexive TA data saturation is not a concept which is supported (Braun and Clarke 2019b). Novice researchers must contend with understanding the varying processes of qualitative data and the specifics within the analytical method they have chosen to use. In doing so they need to understand the terminology and concepts within a particular method to be able to approach and apply this to their thinking and analysis of their data. Remaining current is pivotal, methods can change over time in parallel with researcher's thinking as they develop their analytical craft. As with Braun and Clarke's work as they reconceptualised their TA approach into Reflexive TA to acknowledge the importance of a researcher's methodological positioning, reflexivity and subjectivity in the analytical process of TA. Sharing ways of how to approach data analysis using these different qualitative methods can support novice researchers in developing their understanding and application within their own data analysis. Recognising that these analytical insights are located in a particular moment in time and are likely to develop further over time as researchers develop their craft through experience and application of the method within their data analysis. However, in papers reporting qualitative studies, the 'how to' is often absent in the description of data analysis, resulting in a lack of insight and understanding of how data analysis was undertaken. This can impinge novice researchers in developing their knowledge of analytical processes within qualitative research. This may of course be due to the restrictive word limits offered by journals, but it has led to criticisms in qualitative research in the lack of transparency and understanding in the analytic processes a researcher has engaged with (Noble and

Smith 2015). This does not best support novice researchers or those new to qualitative research in learning 'how to do' analysis. To support addressing this issue, Braun and Clarke have devised guidelines to report and evaluate qualitative research taking a value-based approach which provides flexibility in the creativity of qualitative analysis (Braun and Clarke 2025). This extends their work on the importance of quality in reflexive TA, acknowledging that there is no ',one size fits all' in the analytical approach of qualitative research (Braun and Clarke 2020a). This provides novice researchers with concepts to consider in how they write-up their analysis whilst also providing reviewers with tangible concepts for consideration when reviewing qualitative research.

The aim of this paper is to contribute to these analytical reflections and support novice reflexive TA researchers in developing the craft of this method to benefit their research. This worked example will add to the small body of published examples in understanding the 'how to' of applying this method in practice. My own doctoral research (HP) explored parental treatment decision-making when their child had relapsed or refractory neuroblastoma, a poor-prognosis childhood cancer. The complexity of the analysis involved multiple data from different timepoints which were analysed within and across each treatment decision timepoint to explore how parent decision-making changed over time.

The paper begins by detailing my study briefly, my methodological approach and positioning. The main body of the paper provides extensive details of the application of reflexive TA reflecting on the challenges, iterative process and contextual decisions made of and towards data analysis. Reflexive diary excerpts provided in boxes throughout the paper show my thinking of and towards the data in developing the analysis. Diagrams show the development of the analytical process throughout the analysis.

The research study

Neuroblastoma is a childhood cancer predominately seen in children under 5 years of age. Approximately 100 children are diagnosed in the United Kingdom each year (Public Health England 2021). Fifty percent of these children will have a high-risk disease which is associated with poor outcomes (Morgenstern et al. 2016). For these children, if the disease relapses or becomes refractory where it does not respond to initial treatment, there is no standard treatment protocol (Schrey et al. 2015) meaning there is no superior treatment that has been found to be most beneficial. Treatment options include clinical trials, 'off-protocol' treatments and experimental therapies. With no standard treatment there is no clear treatment endpoint for relapsed or refractory neuroblastoma. Due to this, parents become involved in making repeated treatment decisions for their child in the pursuit of cure. Parents continue to make repeated treatment decisions over months or years depending on their child's response to treatment, toxicities of treatment and treatment availability. Research to date on parent treatment decision-making in children's cancer has focused on single decisions, such as opting to participate in a clinical trial or withdrawing care (Bluebond-Langner et al. 2007; Hinds et al. 2009; Mack et al. 2019). How parents make repeated treatment decisions over time remains unexplored. This study aimed to address this gap in knowledge.

My study sought to answer the following research questions: 1) identify, describe, explain and explore how parents make treatment decisions when their child has relapsed or refractory neuroblastoma; 2) Investigate whether parent decision-making changes over time and if so how?; 3) explore, explain and describe the role of emotion in parent treatment decision-making when their child has relapsed or refractory neuroblastoma.

Data were collected using semi-structured interviews conducted virtually (due to the coronavirus pandemic) with parents after they had made a treatment decision. Parents could be re-interviewed to explore decision-making over time if they made more than one treatment decision during the period of the study. Eighteen parents participated across single and dyad interviews. Interviews consisted of parents who had made between one and six treatment decisions and three parents were interviewed twice.

The Patient Public Involvement (PPI) group supported the study. The group comprised five parents and carers with lived experience of caring for their child with relapsed or refractory neuroblastoma. Data analysis was supported by two parents in the group. The purpose of PPI was to talk through each theme in relation to the codes and data to help with interpretation and shared meaning within and across themes.

Methodology

A researcher's methodological stance informs data collection and analysis. Reflexive TA is 'methodology flexible' with different orientations which hold their own assumptions and approaches (Braun and Clarke 2022b). Each orientation holds conceptual and theoretical thought processes with different approaches to the construction of language (theory), knowledge (epistemology) and reality (ontology). The approach taken in my research study was fully embedded in the qualitative paradigm providing flexible, interpretative, and subjective/reflexive processes; referred to as 'The Big Q' qualitative approach (Braun and Clarke 2022b).

Ontology

My study took a relativistic approach where social reality is dependent on individual interpretation and knowledge (Braun and Clarke 2013). It is a person's interpretation that made something a reality. As a result, multiple social realities exist (Crotty 2003) which are constructed by individuals (Braun and Clarke 2013; Burr 2015). This reality is affected by time and context (Rees, Crampton, and Monorouxe 2020). Social reality requires knowledge to be sought from multiple perspectives from those with lived experience to provide an in-depth understanding of the phenomenon being researched.

In relation to my research study, parent knowledge was their interpretation of their values, preferences, experiences, beliefs, perceptions, and assumptions in relation to making decisions. Reality was created by each parent, resulting in multiple realities existing which can differ between parents who care and support the same child. As reality is not fixed in time, parent interpretation and knowledge can change over time based on their developing knowledge and experience of making repeated treatment decisions. The context in which parents made decisions had the potential to change based on the availability of treatments, their child's clinical condition, and their own personal growth through the experience. My clinical experience suggested these multiple realities are diverse but also share commonalities.

Epistemology

Cognitive constructivism was the epistemological approach taken. Individuals use cognitive processes to construct knowledge and meaning making of their social and psychological worlds (Talja, Tuominen, and Savolainen 2004; Young and Collin 2004). The individual makes sense of their reality and attaches personal meaning to the knowledge constructed (Talja, Tuominen, and Savolainen 2004). This construction of knowledge produces 'mental models' of the world consisting of scripts, schemas, and knowledge structures (Talja, Tuominen, and Savolainen 2004). These 'mental models' evolve as new information is received. My research study found as parents made more treatment decisions over time, parents gained experience and knowledge through repeated decision-making.

Cognitive constructivism acknowledges an individual's information needs can be affected by their current situation, emotional and cognitive states (Talja, Tuominen, and Savolainen 2004) and uncertainty (Kuhlthau 1993). For parents, there is uncertainty on treatment outcomes and toxicities, with decisions being made in a state of high emotion due to their child's illness. Parents experience emotions such as grief, sadness and shock when their child has relapsed disease (Hinds et al. 1996) and involvement in decision-making can be emotionally challenging (Polakova et al. 2024). Uncertainty and emotions have the potential to affect parent cognitive processes and how they construct and utilise knowledge.

During parent interviews, the dialogue was co-constructed and located in that moment of time. My orientation acknowledged parents may have spoken differently if interviewed at a different timepoint, or by a different researcher who may hold alternative subjectivities and positionalities. For example, if a parent was interviewed at a time when their child was tolerating treatment, they may have portrayed a more positive treatment decision-making experience compared to if their child was unwell and suffering as a result of treatment. The dialogue was contextualised depending on their current situation. This was important as the social reality parents constructed was likely to change over time relevant to their experiences and interactions since their last treatment decision.

Reflexivity and positionings

Reflexive TA is interpretative and subjective (Braun and Clarke 2006, 2019a, 2022b). The researcher plays an active role in analysing data. This active role provides sense-making of the data which is influenced by the researchers' reflexivity and positioning in relation to the research subject and population being studied.

I am a white English heterosexual female in my late thirties. I am a children's cancer nurse, and this research study was developed out of my clinical experience supporting parents whose child had relapsed or refractory neuroblastoma. I have knowledge on the treatment options available and the wider context of neuroblastoma in relation to treatment side effects, how these can be managed, and how the disease can manifest over time. My positionality aided the research process as I was able to quickly build a rapport, trust and understanding with participants as I had knowledge of the context parents spoke about. However, as a nurse I often want to 'fix things' so I was consciously aware not to offer suggestions and to listen without judgement. My positioning had the potential to make assumptions about and towards the data as I was so familiar with the circumstances which parents talked about emphasising the importance of reflexivity throughout the research process.

Reflexivity involved a field notes journal to record details from each interview including my thoughts, emotions and judgements. For example, my perception of my relationship with participants and their responses, interview context whether there were any interruptions or distractions which influenced the interview, responses I should have explored further, ideas I had on the interpretations of the data and my interviewing skills. Throughout data analysis, I kept a reflexive journal which acknowledged my assumptions (what I thought of and towards the data), positionality (whether my thoughts related to my clinical nursing experience and if so how, and why), and expectations (pre-conceived ideas and notions related to my clinical experience) bought to and from the data (Braun and Clarke 2022b). This helped me to critically engage with the data on a deeper level exploring interpretative participant meaning. Keeping field notes and a reflexive journal helped discussions of my analytical developments with my supervisors (MM, ASD and FG). This provided opportunities for reflexivity and to explicitly acknowledge my positionality, to critique how I approached the data and the influence my assumptions and positioning had on my analysis.

Inductive and deductive approaches to analysis

Data analysis incorporated inductive and deductive approaches due to analysing multiple datasets of different timepoints to investigate whether parent decision-making changed over time.

Inductive

An inductive approach involved open coding across the whole dataset (interviews at different timepoints). Data were analysed with relevance to research questions, exploring all aspects related to how parents made treatment decisions, how this decision-making may change over time and the role of emotion in making decisions. An inductive approach was considered appropriate, as the literature within children's cancer has not investigated how parent decision-making changes over time nor the role of emotion in decision-making. To support an inductive approach, I ensured I was present in the moment with the data by removing distractions such as my mobile phone, working in silence, and away from other people, so I was focused and immersed in the participant words.

Deductive

As one of my research questions addressed decision-making over time, a deductive approach was applied from treatment decision timepoint two (T2) onwards, where parents had made more than one treatment decision. A deductive approach enabled me to analyse whether there were differences to how parents made repeated treatment decisions at each decision timepoint. This involved moving iteratively back and forth across codes and themes from previous treatment decision timepoints to explore what was the same, or different, for parents making decisions over time.

From T2 onwards, inductive coding explored what was new or had changed over time. New codes and themes developed at each timepoint, were incorporated deductively into the ongoing analysis of subsequent decision timepoints. Interpretation and write-up of study findings were situated and interpreted within existing research and theory relating to themes and key concepts from the analysis.

Methods of reflexive TA

Reflexive TA was the chosen method for data analysis as it aligned with my methodological paradigm and data collection using semi-structured interviews. The goal was to draw meaning, establish shared patterns of meaning and similarities between and across cases to answer research questions.

Braun and Clarke developed six phases of reflexive TA: 1) familiarisation; 2) coding; 3) generating initial themes; 4) developing and reviewing themes; 5) refining, defining and naming themes; 6) writing up. A researcher engages with these phases opening and knowingly which blend together as the analytical process develops to critically engage with the data iteratively (Braun and Clarke 2020a, 2022b). I closely followed these phases at the beginning of my analytical journey, with the analysis becoming more fluid and iterative as my confidence and experience of the method developed. Braun and Clarke term this as 'once you feel confident about doing TA, such guidelines can be put to one side, maybe occasionally check back with' (Braun and Clarke 2022b, xxix) How I applied the methods of reflexive TA to my analysis can be seen in Figure 1.

NVivo Release 1.0 (Lumivero 2020) was used to store interview transcripts, support initial familiarisation (step one), organise my codes with the associated data. Analysis of the data, developing codes and themes was done using hard copies as this was easier to work with than computerised software. Hard copies provided flexibility and creativity by being able to move codes and associated data around for sense-making and looking at shared patterns of meaning.

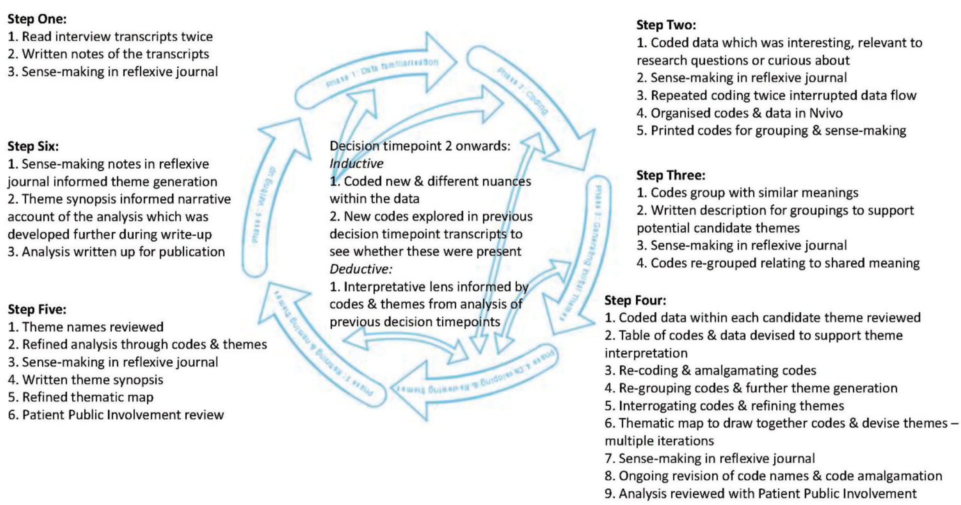

Figure 1. Applied approach to reflexive thematic analysis in this study.

Phase one: familiarisation

Data familiarisation involves the researcher developing thorough knowledge of the dataset, critically engaging with the transcripts as data and writing notes related to thoughts on the dataset (Braun and Clarke 2022b). I did this through immersing myself in the interview transcripts reading and re-reading which allowed me to actively become familiar with the data. I was then able to write reflexive notes on what I thought was happening in the data to make meaning of the phenomenon (parent treatment decision-making) being researched. I read the transcripts twice, leaving time between each reading to absorb and reflect. The purpose was to understand participants' stories and reflect on their potential meaning. After reading the transcripts I uploaded these to NVivo and used the annotation feature to make notes on my thoughts and questions of what I believed to be happening in each transcript. Transcripts and annotations were then read together to provide further clarity on participant meaning. I recorded 'sense-making' notes in my reflexive journal about these annotations for each treatment decision timepoint. See (Box 1). for my reflexive thoughts on treatment decision timepoint one (T1).

Box 1. Reflexive journal sense-making notes for parents who had made one treatment decision.

My positionality as a nurse working with this parent population in clinical practice is that parents may or may not have an inkling that their child has relapsed or has refractory disease. This can impact how they receive this information from their child's medical consultant and as a result how they engage with decision-making at this timepoint. Concepts I noticed in the data were:

- There is a focus on the relationship with their child's medical consultant which seems important in their involvement in decision-making. Not explicitly mentioned but trust, knowledge and information sharing feel important to parents
- Parents are concerned about the impact on the wider family ie: siblings, the need for family time
- How parents manage the situation – some parents speak of being in the moment and not looking at other treatment options whereas others want to explore treatment options and make future plans
- Understanding or awareness of their child's prognosis appears varied at this timepoint which could be linked to how parents manage the situation
- Some parents do not want to be involved in making decisions which could be linked to their understanding of treatment options
- Parents spoke of not wanting to know information which contradicts the need for information that they also spoke about. This might be linked with their emotion in managing their situation
- All parents spoke of their child's medical consultant making the treatment decision either to 'make a quick decision to start treatment' or 'because I wasn't in the right mindset'

There are contradictions within the data from some of the parents, concerns and worries about the future but also potentially not wanting to 'go there' for fear of what they may be told. The data feels ladened with implicit emotion, managing the news of the diagnosis and knowing that no treatment protocol means decision-making is difficult.

These initial thoughts supported coding and data familiarisation of transcripts at subsequent decision timepoints.

After reading all the transcripts and making notes I summarised my data familiarisation experience in my reflexive journal (Box 2).

Box 2. Reflexive account on data familiarisation.

"Familiarising myself with the data felt good. I was able to read interview transcripts to hear their stories which gave a sense of shared meanings between parent experiences and where narratives differed. There is a strong sense of parents wanting to be involved in decision-making which supports the literature to date within the field of parental treatment decision-making in children's cancer. However, their involvement feels more intense possibly due to the poor-prognosis and the want/need to save their child's life.

Reflexive TA is not about how common an issue is, and I need to immerse myself further through coding to understand each transcript individually before interpretating and making assumptions. I am starting to think about the latent and semantic coding, I am focused on the participant meaning as I can relate to this from my clinical experience (positionality) but how does latent implicit meaning develop? Will coding help with this? Am I overthinking it?"

Moving onto coding will help develop my analytical thinking, spending more time immersed in the data.

Data familiarisation provided me with a thorough insight into each transcript to understand participants' stories. At the end of this step, I knew the transcripts extremely well and could take a segment of data from a transcript and summarise participants' stories without needing to read the whole transcript. As reading and re-reading the transcripts and annotations did not provide any new insights, I commenced coding.

Phase two: coding

Coding is the process the researcher engages with to generate codes and code labels which are the 'building blocks' of the analysis (Braun and Clarke 2022b). Coding and re-recoding was an iterative process that moves back and forth throughout the dataset to explore diversity and patterning meaning at each treatment decision timepoint. During coding of interview transcripts, I reviewed the corresponding field notes to provide further context and made additional notes on the transcript to support understanding. Notes related to commonalities and differences between interviews, potential implicit meanings and speculations. For example, as coding progressed through the T1 interview transcripts, although all parents were given treatment options by their child's medical consultant the final decision was made by the healthcare professional. Speculation of this related to parent emotional processing of being told their child had relapsed or refractory disease and requiring time to absorb it before becoming involved in the decision-making process. As coding progressed, codes were refined across the interview transcripts over several iterations. I coded segments of data which were interesting, or potentially relevant to the research questions or where I was curious about them. As I become more confident in my approach my coding became more focused in relation to my research questions, resulting in less codes overall. Codes provided a summary of the meaning of data staying close to words explicitly spoken by participants (Braun and Clarke 2022b). At this stage codes were descriptive (semantic), capturing obvious surface meanings in the data staying close to the language used by participants rather than interpretative (latent) which captures implicit or conceptual meaning (Braun and Clarke 2022b). Coding started semantically as I found it difficult to move away from what participants had said for fear of misunderstanding or misrepresenting their experiences. Table 1 shows a selection of the data with code labels.

Coding required a systematic and robust process (Braun and Clarke 2022b). I dated each entry in my reflexive journal to support the process of my analytical developments. This was important as I had periods away from data analysis and it provided a robust audit trial of my thinking process and progress which I could articulate to my PhD supervisors. My reflexive journal incorporated my thinking on what was happening in the data, potential reasons why I thought that and a summary of what I had done that day to re-read which helped me re-engage with my analysis after a period of time away.

Interviews were assigned to a number in the order they were conducted. Coding started in ascending number order of participant transcripts within each decision timepoint. Subsequent coding started in the middle of the dataset descending to one and then working down from the last transcript to the middle. This approach interrupted dataset flow familiarity and allowed for even refinement across the dataset as further insight was gained (Braun and Clarke 2022b). This approach meant subsequent coding did not start with the most recent transcript I had been working on and reduced the potential for disengagement in the coding process through familiarity. I worked systematically through coding the dataset three times informed by my field notes and notes made within each transcript in the first round of coding. I approached coding curiously, recognising the need to hear and report all experiences even if they did not align with my clinical experience. As my experience and confidence grew, my coding became a combination

Table 1. Selection of data from the first treatment decision and code labels.

Data	Code labels
"When you're desperately looking for any information, or you just feel like you just need to be seeing things or reading things. Equally, I think we were so terrified of relapse"	Fear of relapse Scared for their child's future The need for information to feel some control Being informed helps with coping
"We couldn't say we made those decisions based on the evidence we have, or the knowledge that we have. It was more hoping for the best, and trusting the people helping us" *"We are hoping that one day they say ***** is cancer free"* *"Hoping that we get the best of the treatment"*	Hoping for a positive outcome Having the best treatment provides hope Limited knowledge and understanding to inform decision-making Reliant on others for help and guidance
*"To have ***** well and out, even for a walk is very important for us I'd rather have ***** playing and enjoying ***** time, rather than be unwell due to treatment"* *"We wanted ***** to get ***** strength back, but still get some treatment, because we knew that if they stopped there and then, that was it finished"*	Maintaining child's Quality of Life is essential Prevent treatment suffering to maximise quality time with child Need to balance child's Quality of Life with treatment Recognising no treatment meant their child would die

of descriptive (semantic) and interpretative (latent) codes. Latent coding is abstract, using a deeper analytical lens where the researcher explores interpretative conceptual meaning to what participants have said (Braun and Clarke 2022b). I reflected on the developments of my coding in my reflexive journal (Box 3).

Box 3. Reflexive journal on coding.

"The coding feels very semantic, descriptive, and close to what the participants have said, and at times I feel I could be paraphrasing. Parents have spoken so openly in the interviews about their experiences, feelings and emotions that I do not want to misrepresent them as I develop my analysis. Prior to undertaking any interviews, I was concerned what parents would share particularly as they did not know me and the interviews were being conducted virtually due to the Coronavirus pandemic. However, I learnt that having the opportunity to speak with someone unconnected to their situation was beneficial to parents – they could speak without feeling judged by friends, family or healthcare professionals and it helped their thought processes towards decision-making for their child. I learnt there was a huge different between being a clinician working with this parent population to being a researcher. Parents shared more in-depth, raw thoughts, feelings and emotions during interviews than I had ever heard parents speak of in clinical practice. Perhaps this was a protective mechanism in clinical practice, but coding made me see there were many nuances both implicit and explicit within the data. For example, the effects on parent emotions having a child diagnosed with cancer and then subsequently being involved in making treatment decisions for their child.

I needed to move away from coding within transcripts to work with codes and associated segments of data in the latter stages of code refinement. I felt nervous about this stage and found it difficult to move away from the transcripts due to a lack of confidence in my skills. To help overcome this anxiety, one of my supervisors (FG) reviewed a coded transcript and did not pick up on any additional concepts which encouraged me to move forward with collating codes and relevant data segments.

To support this, the handwritten codes were created in NVivo and assigned relevant data segments to these codes. Codes varied in length to capture the meaning and interpretation of data. The list of codes was printed from NVivo twice – one copy in its original list format for reference, and one copy was cut up into individual codes which were laid out on a table (Figure 2). This enabled me to physically move codes around, remove duplications, refine codes which had the same meaning or evolve codes as my analytical insights and understanding developed across the dataset. Codes were also grouped for sense making. As a researcher who prefers to work using visual representations, having all codes laid out on a table enabled me to go back and forth iteratively. Using NVivo required scrolling through the code list and I found this method made it difficult to develop meaning-making between codes. Moving between technology and hand copies is supported by Braun and Clarke (2022b) who emphasis engaging with the process in a way that works best for the researcher. Once I had printed out the codes and data segments from NVivo, the rest of my analysis was done using hard copies allowing for flexibility and creativity.

Figure 2. Initial codes from T1.

Codes should provide meaning and diversity which make sense without data attached (Braun and Clarke 2022b). This allowed me to interrogate and clarify codes to ensure meaningful richness. I could see some codes lacked clarity and were unclear. I questioned these codes and revisited the relevant data to refine them (Table 2).

Code refinement developed as more time was spent immersed in the coding process. For example, there were two codes for ‘parent partnership in decision-making’ which contained the same data and were therefore merged. Printing out codes and grouping these helped with refining codes. Codes which had the same meaning were grouped and assigned a new code (Table 3).

Grouping codes showed that some codes contained multiple data from one transcript. For example, codes ‘lack of control in treatment outcomes’ and ‘unknown outcomes make treatment decision-making difficult’ contained the same data. I anticipated these codes would have shared meaning with other codes once I started to generate initial themes. However, I was conscious if I continued to group codes into larger meaning units, there was potential to lose meaning, and the analysis could become narrow and superficial. At this time, I clarified my thinking whether I was ready to stop coding and move onto generating initial themes. I re-read chapters three and four of the Braun and Clarke (2022b) textbook. This enabled me to review and reflect on my analytical approach acknowledging I could move back and forth across the phases to develop my analysis further during phases 3–5. This iterative non-linear process gave me confidence to move onto phase three.

Table 2. Examples of codes which were refined and revised.

Initial code	Questions of the code	Revised code
Ability to ask for help in making decisions	Ask for help from who? What type of help was needed?	Ability to ask consultants for information and clarity
Rarity of disease limits understanding	Understanding of what?	Rarity of disease limits understanding of approaches to treatment
Difficulty in predicting treatment response	Why is it difficult to predict?	Treatment response individual to child

Table 3. Examples of code grouping and revisions.

Revision of initial codes	New code
'Being able to live with decisions made' 'Being okay with decision made' 'Having to live with the decisions made' 'Living with the decisions made'	Being able to live with decisions made
'Advocating to commence treatment with professionals' 'Urgency to commence treatment by parent' 'Parent need to start treatment'	Parent need for treatment to be administered in a timely manner
'Awareness that relapse disease is harder to treat' 'Recognition treatment needs differ at relapse' 'Recognising relapse is more complicated than front-line treatment' 'No superior treatment for relapse'	Recognising relapse is harder to treat with no superior treatment
'Lack of control in treatment outcome' 'Unknown outcomes make treatment decision-making difficult'	Treatment decision-making complicated by lack of control on outcomes
'Terminology as a barrier to understanding treatment' 'Terminology as a barrier to understanding treatment options'	Terminology as a barrier to understanding treatment

Phase three: generating initial themes

Generating initial themes requires the researcher to engage with coded data analytically across the dataset to explore patterns of shared meaning underpinned by a central organising concept and clustering these codes together into potential candidate themes (Braun and Clarke 2022b). I started clustering codes into potential candidate themes which I thought had shared patterns of meaning in relation to the research questions. I grouped (clustered) and re-grouped codes under themes to explore shared meaning writing descriptions of these candidate themes to support analytical development and my reflexivity.

Having spent considerable time refining codes in phase two, I was very familiar with the coded data which enabled me to group codes with similar meanings easily. To ensure I identified shared patterns of meaning relevant to the research questions, I always had the research questions to hand for reference. For example, 'time' appeared to be a factor in how parents made treatment decisions. This related to time spent with their child or time lost if the treatment did not work. Potential candidate themes were developed out of this grouping and to support my initial analytical thinking I wrote a description on what each candidate theme captured. Themes were: ‘The concept of time’; ‘The psychological yo-yo’; ‘Treatment impacts on child’; ‘Support from a community’; ‘Professionals as leader or follower’; ‘Emotional processing of the situation’; ‘Parent research and information gathering as a necessity’; and ‘Miscellaneous codes.’ Figures 3 and 4 show codes for the candidate themes ‘Treatment impacts on child’ and ‘Professionals as leader or follower’ with a written description for each theme. I documented my thinking on developing these potential candidate themes in my reflexive journal (Box 4).

Box 4. Reflexive journal on generating initial candidate themes.

‘With the description of the candidate themes I feel like I am summarising the codes and not providing my analytical interpretation on what is happening. I am also struggling to insightfully name the themes and feel these are not analytical enough. I recognise that these are initial theme names which will change as my analysis develops. I am hoping further iterative engagement with the themes and codes will provide clarity and direction. Discussions with my supervisors and members of the PPI group will also support critical thinking of and towards the data.’

I met with my supervisors to discuss candidate themes and written descriptions. This provided an opportunity to talk through my analytical approach so far, questioning why and how I have clustered codes into shared meanings across the dataset. During this discussion, I reflected on the work required to continue to develop my analysis which meant spending more time engaging with codes to identify shared patterns of meaning. Time to critically reflect and talk through my analysis with more experienced researchers was an essential part of my analytical development and self-reflection within my analysis.

After this meeting, I returned to the Braun and Clarke (2022b)textbook and re-read the developing and reviewing themes section (phase four, p.97). I then reviewed codes within each candidate theme which showed I had collated positive and negative codes which supported or refuted the theme. For example, ‘Professionals as leader or follower’ included codes where professionals led decision-making and codes which relinquished control to parents. I could see themes were summarising what the codes related to. For example, all codes relating to professionals were collated under the theme ‘Professionals as leader or follower.’ These themes felt like domain (topic) summaries: I had brought codes together which related to the same topic as opposed to codes which shared the same or similar meaning. Organising data into domain summaries rather than analytical themes is a recognised pitfalls in reflexive TA (Braun and Clarke 2022c) and novice qualitative researchers are susceptible to this. I recognised my error after having organised my candidate themes and written descriptions through discussing my analysis with experienced qualitative researchers (FG, MM) and re-reading chapters of the Braun and Clarke (2022b) textbook. I learned that the analytical process could mean

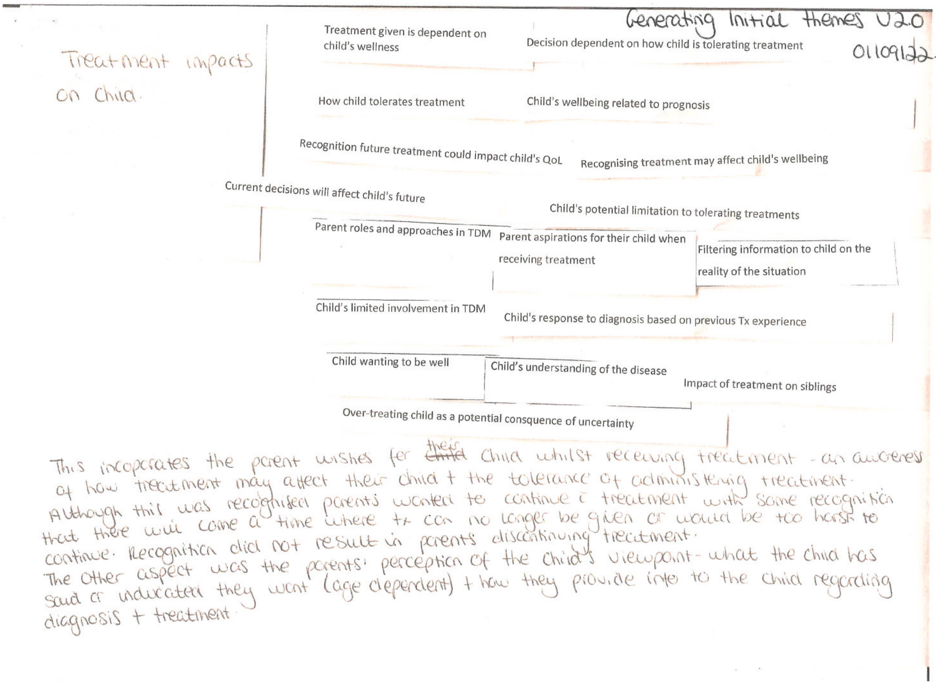

Figure 3. Potential candidate theme for 'treatment impacts on child'.

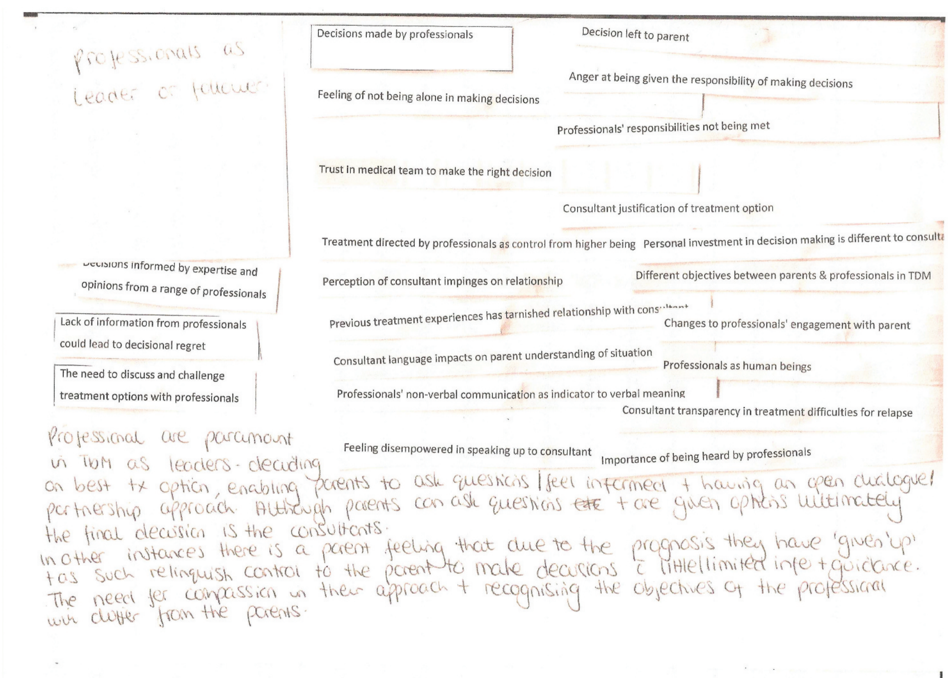

Figure 4. Potential candidate theme for 'professionals as leader or follower'.

disappearing down rabbit holes which can strengthen your understanding of the analytical process and the need for deeper critical engagement with the dataset.

Reviewing themes shifted my analytical thinking about codes. For example, the code 'previous treatment experiences have tarnished relationship with consultant' was within the theme 'Professionals as leader or follower' as it related to professionals. Reviewing this code on a semantic level, it related to a parent's relationship with the professional, however interpretation of the code saw this was how that relationship shaped and influenced parent decision-making. Figure 5 shows the analysis after I had separated positive and negative codes from the initial candidate themes. For example, the code 'previous treatment experiences have tarnished relationship with consultant' was grouped with 'justifying previous decisions made', 'decision informed by previous treatment experiences', and 'previous treatment experiences affect current concerns on side effects'. These four codes were grouped under the candidate theme 'past experience and knowledge influences current decisions.' This helped me to think more interpretatively about the shared meaning between codes to develop further analysis. I immersed myself iteratively, going between codes to review and refine candidate themes. This engagement supported my analytical thinking establishing shared meaning between the codes with implicit conceptual meaning which helped to further develop my candidate themes.

Two processes contributed to developing the analysis further: 1) I became more experienced as my analysis developed (which supports Braun and Clarke's positioning on semantic and latent coding) spending immersed time with my themes and associated codes; 2) I had periods of reflexivity and time away from data to question and conceptualise my thoughts, assumptions, and ideas of the analysis which I recorded in my reflexive journal. Analytical development requires 'headspace' with time away from the data to allow for inspiration and insight to develop (Gough and Lyons 2016). From this stage forward the analysis became increasingly complex with numerous rounds of iterations moving between phases 3–5 illustrating further the need for a flexible approach and not following the phases in a linear manner.

Phase four: developing and reviewing themes

This phase builds on the analytical process of the initial theme development through re-engaging with all coded data extracts and the complete dataset (Braun and Clarke 2022b). This was an iterative process of reviewing the coded data within each candidate theme with continued theme revision to support shared patterned meaning. Theme revision involved an iterative back and

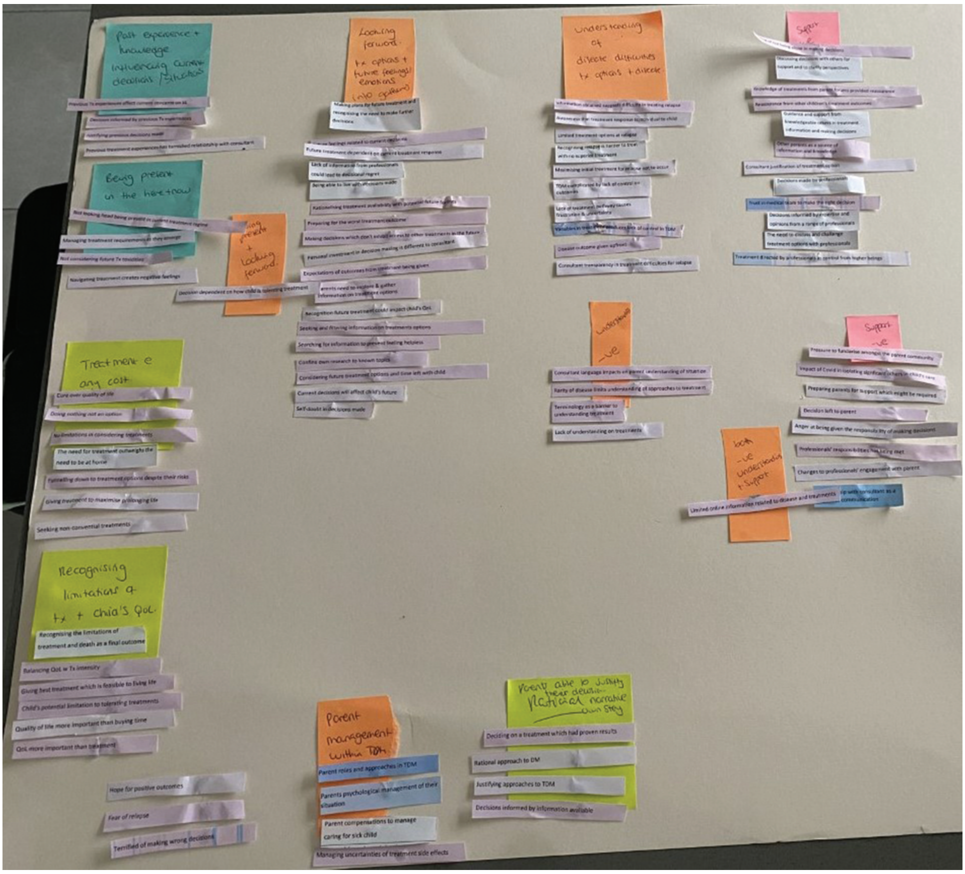

Figure 5. Analysis after reviewing initial candidate themes.

forth process incorporating phases two and three with re-coding and refining codes to be able to generate themes, review and refine them through ongoing engagement with the coded data. Initial interpretation of the connections between themes were considered to support the overall narrative of the data.

Review of codes

Each code and associated data were reviewed within the candidate themes (Figure 6) to see whether themes captured the complete essence of the data within that theme. Hard copies of data made it easier to see where data had been assigned to multiple codes, moving back and forth between codes easily. I was able to see whether a theme was centred on the views of a single participant which is not the basis for a theme in reflexive TA. The theme should have shared patterned, meaning across the dataset (Braun and Clarke 2022b). Where codes were across more than one theme, I reflected on the method of reflexive TA. Codes can inform more than one theme if elements of the code fit differently (Braun and Clarke 2022b) which supported analytical

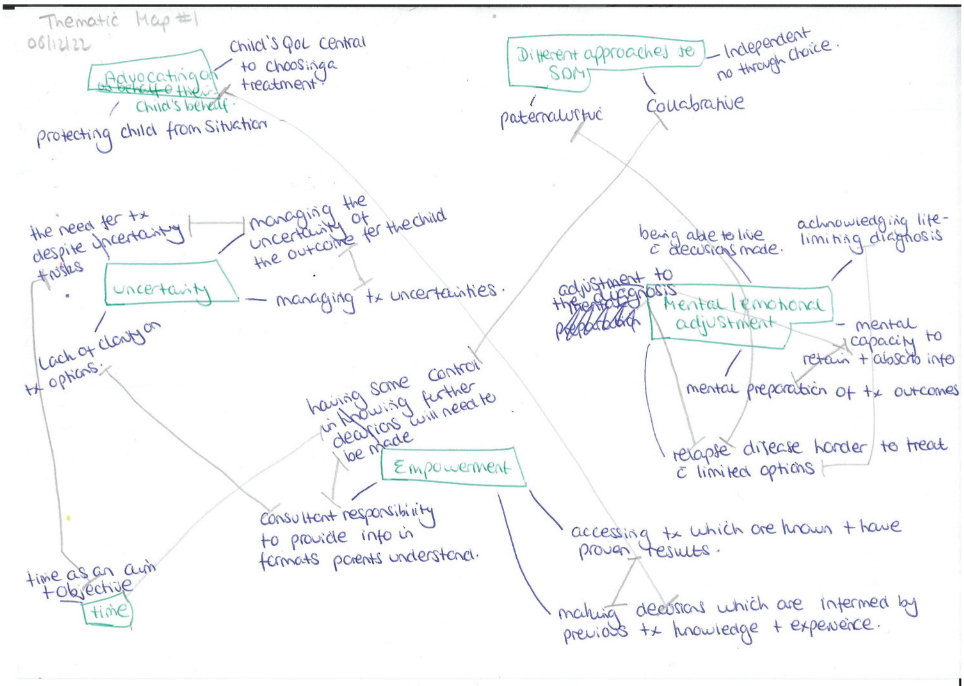

Figure 6. Thematic map version 1.0.

development of shared patterns of meaning. I felt overwhelmed at this stage and wrote in my reflexive journal (Box 5).

Box 5. Reflexive journal on reviewing themes.

'There are themes which contain codes with data that is assigned to more than one code within that theme. There is also duplication of data across codes in different themes which could support patterned meaning with connections between themes. However, themes are meant to have boundaries. Trying to build a narrative within each theme and across the dataset has left me unsure how relevant it is to have data assigned to multiple codes within and across themes. The analysis requires further refinement. I need to discuss with my supervisors to help my critical thinking on how to develop my analysis further.'

I met my supervisors to discuss the themes in their current format. Our discussion made me see that a theme could have multiple components, even opposite viewpoints if they shared patterned meaning. This required further interpretative analytical thinking. We discussed data which were assigned to multiple codes and agreed I should review these codes and the extracted data to think about the interpretation and shared meaning across and within each theme. Importantly, we discussed the need to review codes in relation to research questions to stay focused.

To support my analysis, I devised a table with the research questions and assigned the relevant codes to each question. The table contained five columns: 1) codes; 2) grouping of codes which captured the meaning of the codes within column one; 3) 'what does this mean?' for interpretation of data and theme development; 4) column with data extracts which supported the code; 5) column for the literature which supported or refuted my findings to support my analytical write-up contextualising, and interpreting findings with research and theory as appropriate. For data where parents had made one treatment decision, two research questions were addressed: identify, describe, explain, and explore how parents make treatment decisions when their child has relapsed or refractory neuroblastoma (research question 1); explore, explain, and describe the role of emotion in parent treatment decision-making when their child has relapsed or refractory neuroblastoma (research question 3). The third research question investigated whether parent decision-making changed over time and if so how (research question 2) which was not included within this dataset. These parents had only made one treatment decision at the time of interview therefore they had no prior experience of making treatment decisions to analyse how their decision-making changed over time. Table 4 shows an extract of code and theme development table for T1 relating to research question one.

Re-coding and refining codes

Through reviewing codes and associated data within the candidate themes, I began to see where codes shared patterned meaning and where further analytical development was required. For example, 'hope for positive outcomes', 'fear of relapse', and 'terrified of making the wrong decisions' were codes which reflected parent thoughts when making decisions and the emotional capacity to manage decision-making. I refined codes which shared the same or similar data after spending considerable time reflecting on their shared meaning. Revisiting data within each code helped explore my thinking. For example, the code 'self-doubt in decisions made' and 'future feelings related to current decisions' expressed managing the uncertainties of treatment. I captured my thoughts in my reflexive journal (Box 6).

Box 6. Reflexive journal on refining codes.

'I am not sure whether the new codes are actually themes. If so, the information I have written in the "what does this mean?" column could be seen as sub-themes. Braun and Clarke stipulate that too many sub-themes are suggestive that the analysis is under-developed. I think at this stage the aim is to revisit the codes and associated data to engage deeper with these to explore shared meaning. Perhaps I am trying to get ahead of myself and need to have confidence in my approach and take time to critical reflect and let the analysis develop.'

This process was not linear and required re-engagement with the original codes and associated data from phase two and three to develop my analysis further. I refined my codes and re-grouped these to generate themes which had shared patterns of meaning which provided further insight into data interpretation. For example, a concept within the research question one was around who takes the lead in making these decisions. Codes which related to healthcare professionals leading in decision-making were re-coded to 'paternalistic decision-making'. Codes relating to a lack of direction and decisions being left to parents were re-coded to ‘independent decision-making not through choice.’ Codes which related to guidance and discussion in the decision-making process were re-coded to ‘collaborative partnership in decision-making.’ These three aspects of decision-making shared meaning with the central organising concept of ‘different approaches to shared decision-making.’

I had not anticipated needing to iteratively move back and forth as much to generate, develop and review these themes which took considerable time, patience, interpretation, and critical thinking and engagement to develop the analysis. This iterative non-linear process I engaged with further illustrates the need to be flexible in the analytical process to produce a rich nuanced narrative of the analysis.

Generating themes

I drew a thematic map (Figure 6, Version 1.0) of themes to visualize the codes which made up a theme and the connections between these. Thematic maps can support a researchers’ thinking on how themes stand in their own right, how themes relate to one another, and from this start to consider the overall narrative of the story (Braun and Clarke 2022b). On my thematic map, arrows indicated where codes informed other themes as the complexities of this decision-making were interrelated. Through this process, I could see some codes did not fit, or I was drawing codes together under a theme that did not capture the shared meaning of the codes. The thematic map (Figure 6, version 1.0) was shared and discussed with two members of the PPI group. Given the sensitivity of the research subject, I was mindful that interpretation and theme names may cause distress and disengagement with the research within the parent community. Therefore, having PPI involvement was imperative. Talking through themes provided further clarity on codes which had shared meanings across and within themes and identified codes which could be refined. For example, the code ‘relapse disease harder to treat with limited options’ parents questioned what ‘harder to treat’ was being compared with. This code was refined to ‘recognising relapse disease is harder to treat compared to initial diagnosis.’ Both parents liked the words empowerment, adjustment, and uncertainty within theme names. The theme ‘time’ required further exploration of the different components of time and what this provided parents. For example, time was seen within other codes, such as ‘the need to explore and know there are other treatment options available’, which interpretatively provided more time with their child acknowledging that their child could die and time to explore further treatment options. Following the meeting the thematic map was revised to explore shared meaning of codes across the themes, which could be refined or regrouped (Figure 7, version 2.0). Miscellaneous codes were revisited to incorporate codes which supported analysis. For example, ‘managing treatment requirements as they emerge’ supported the code ‘being present in the here and now.' The code 'asking professionals for information when able to acknowledge the relapse situation' supported the code 'cognitive capacity to absorb and retain information.' These codes were incorporated into the analysis.

A strong narrative was being created, but ongoing theme revision was required. To support my interpretations and analytical thinking I discussed my analysis with my supervisors. Their feedback suggested I focused on what each theme was contributing to the narrative. Identifying the boundaries of each theme and the connections across the themes which can be achieved through thematic mapping. To support my interpretation of my themes, it was suggested I write a synopsis of each theme to support my analytical thinking and clarification towards theme revisions.

Interrogating and refining themes

Codes within each theme were reviewed with associated data again to continue my analytical thinking and interpretation. To support refining themes, I looked at connections between codes and data segments within and across themes. There were codes under the theme uncertainty which related to each other: 'uncertainty of treatment and risks', 'uncertainty of treatment side effects' and 'uncertainty of the child's outcome' which contained some same data segments. These codes were refined into one code 'uncertainty of treatment outcomes, side effects and risks.' I re-read sections of the Braun and Clarke textbook (2022b) on

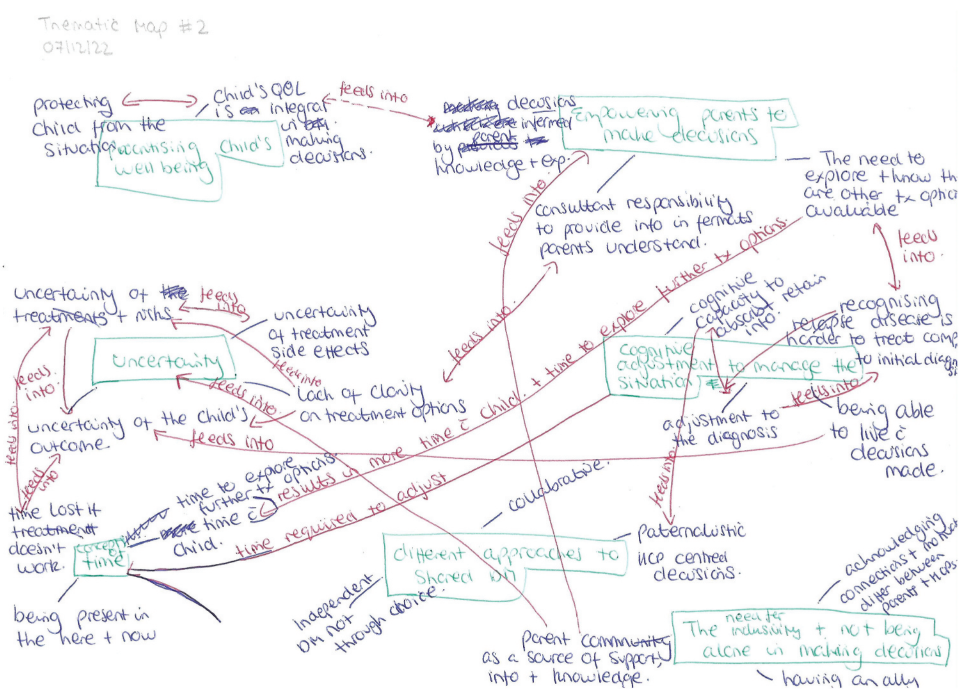

Figure 7. Thematic map version 2.0.

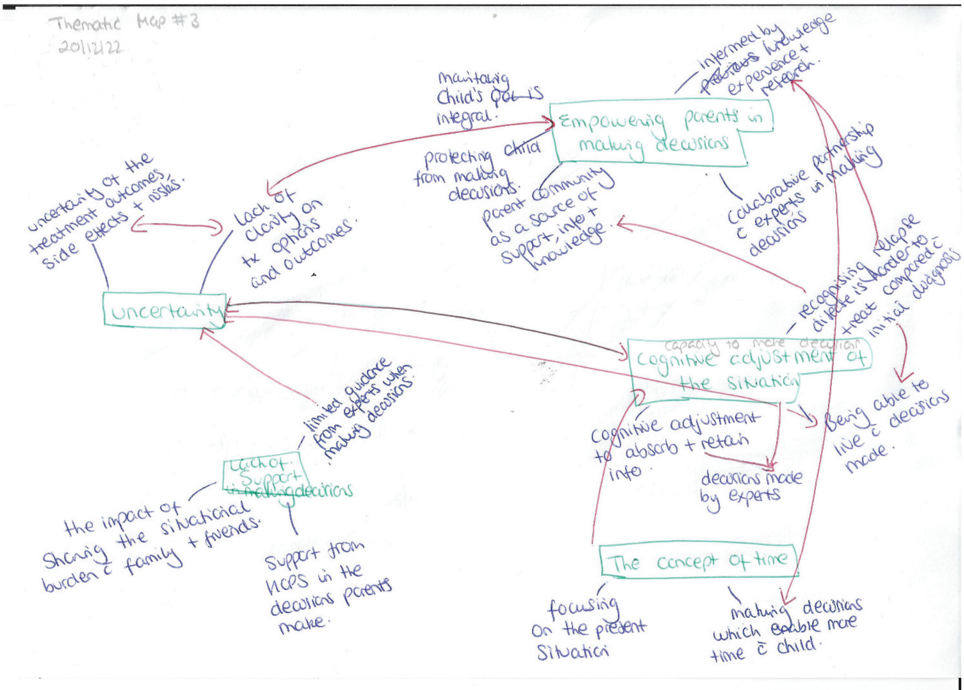

Figure 8. Thematic map version 3.0.

phase 4in particular, developing and reviewing themes. This helped with confidence in letting go of codes and themes which did not fit the narrative of the analysis and answer the research questions. Through reviewing the thematic map version 2.0, I was able to see connections between codes which could be refined and regrouped, developing a more cohesive visual (Figure 8, version 3.0).

Phase five: refining, defining, and naming themes

This involved further development of the themes and analytically refining the analysis to support writing a cohesive structured narrative in phase six (Braun and Clarke 2022b). Practically, I undertook further refinement of the analysis through reviewing and regrouping codes, revising theme names, revising thematic maps and writing a synopsis of each theme to bring together a cohesive narrative of the data.

I started with re-reading Chapter 4 of the Braun and Clarke textbook (2022b) on naming themes. Theme names felt particularly important as this is the first interaction readers have with the analysis, and I wanted them to be meaningful. The theme 'uncertainty' needed further exploration as one-word themes do not capture analysis in relation to the dataset (Braun and Clarke 2022b). The word uncertainty is the topic but did not provide my analytical take on what was happening within the dataset related to uncertainty. I revised this theme to 'Uncertainty of the treatment risks, side effects and outcomes.' The review of theme names also led to further regrouping of codes. For example, the code 'limited guidance from experts when making decisions' and 'support from healthcare professionals/experts when making decisions' were grouped and recoded to 'the need for guidance and support from experts when making decisions.' The theme changed from 'lack of support in making decisions' to 'the burden of making decisions in isolation' to capture consequences when guidance and support is lacking from healthcare professionals. I wrote a synopsis of each theme and during this process wrote in my reflexive journal (Box 7).

Box 7. Reflexive journal on theme synopsis.

'As I am writing the synopsis of each theme, I wonder whether this is a summary of what is happening within the theme as opposed to my interpretation. Should the synopsis include both a theme summary and my interpretation of what is happening which supports the narrative of the analysis? During this writing I can see connections between themes and am questioning how I have bought the analysis together and my interpretation of it.'

I was unable to write a synopsis for the theme 'empowering parents in making decisions' as the theme name and content did not feel right. My concerns related to codes, which made up this theme and whether 'empowering' was the correct terminology which showed how parents were involved in the decision-making process. The thematic map was refined (Figure 9, version 4.0) and the synopsis of each theme was shared with the two parents from the PPI group for discussion. This helped clarify the word 'empowerment' which parents described as '*a positive word being used in a negative context.*' That is, parents do not want their child to be sick and therefore do not want to be involved in making decisions but are involved for the sake of their child. Another aspect I was concerned about was the theme 'the burden of making decisions in isolation' as I felt there were shared meanings between this theme and the theme 'empowering parents in making decisions.' The discussion with parents extended my analytical thinking towards these themes and I could see shared meaning related to being an advocate for their child and how this manifested within decision-making. This critical discussion of theme names further refined the thematic map (Figure 10, version 5.0) and theme synopses. Themes had a boundary, but also a connection with other themes, providing a cohesive narrative of the analysis in telling the story of how parents made their first treatment decision. Shared meaning of each theme supported the methodological approach acknowledging the multiple realities of parent treatment decision-making. The thematic map (Figure 10 version, 5.0) and theme synopses were shared with my supervisors for discussion. We discussed my hesitancy towards the theme name 'being an advocate in the decision-making process'. I had not defined the meaning of 'advocate', and this did not capture

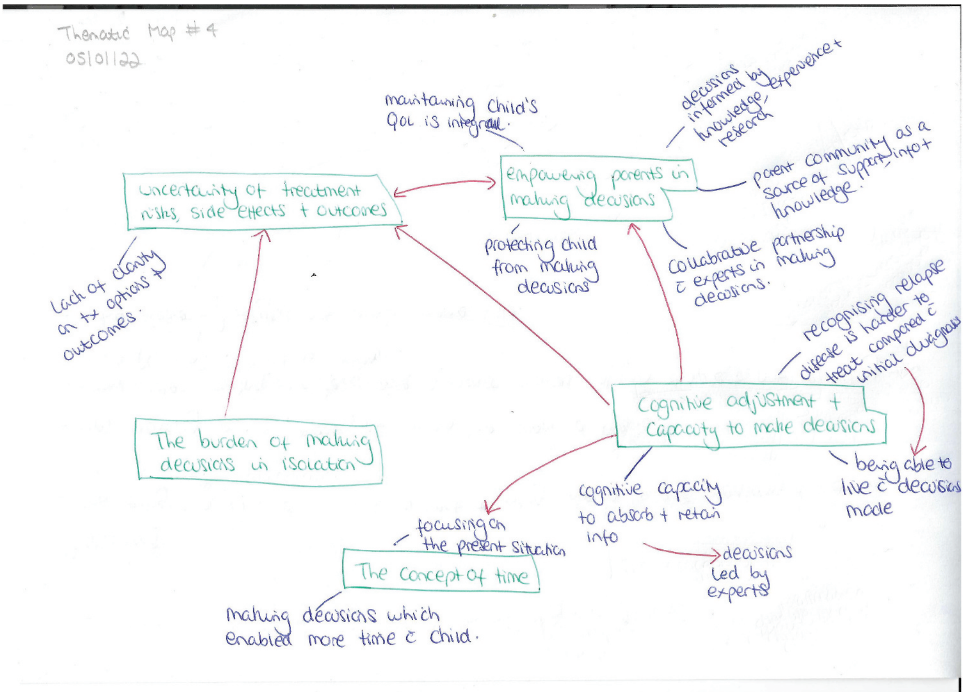

Figure 9. Thematic map version 4.0.

shared meaning across the data as for some parents this decision-making was put upon them. The word advocacy suggested parents wanted to make decisions. I therefore wrote down several options 'being a part of the decision-making process', 'having a voice in the decision-making process' and 'being involved in the decision-making process'. At this stage of analysis, I felt that being a part of the decision-making process' was a neutral theme to encompass the varying realities of involvement for parents.

I shared this narrative with the wider PPI group for feedback. Group consensus of the narrative was that it was relatable to parents/carers who had been involved in making treatment decisions for their child and the human element to decision-making was interwoven strongly throughout. Having spent considerable time immersed in the analysis of parents who had made one treatment decision, which was the largest dataset, I felt confident I had a strong narrative ready to move onto the analysis of subsequent transcripts. This started with parents who had made two treatment decisions.

Analysis of transcripts from treatment decision timepoint two onwards

Analysis of subsequent treatment decisions was approached in the same systematic way as described above. A deductive analytical approach was incorporated from this timepoint onwards. Analysis started with a dataset at

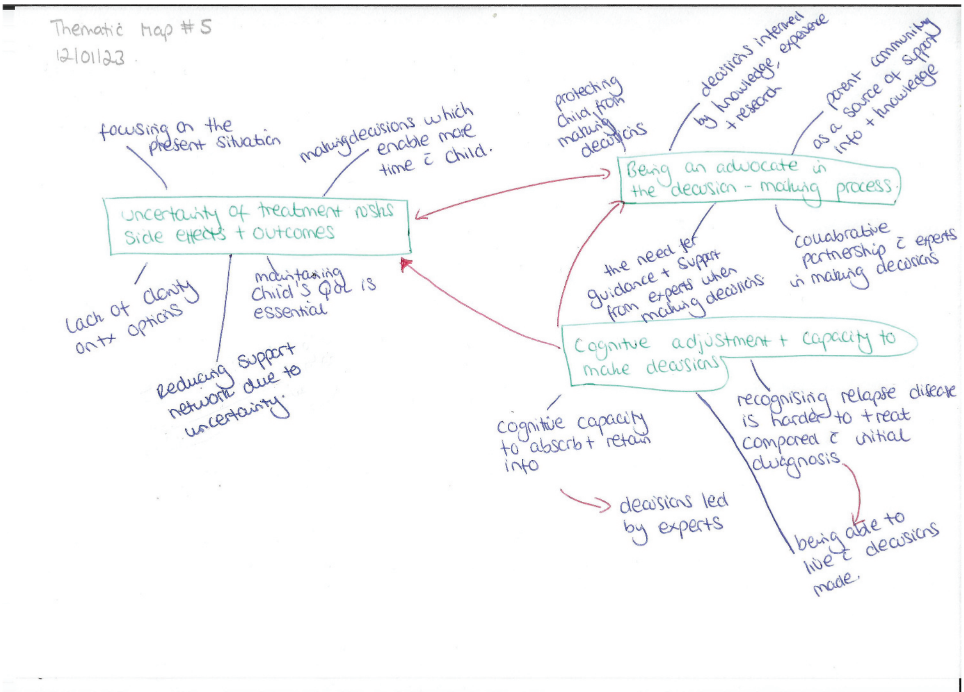

Figure 10. Thematic map version 5.0.

T2 for parents who had made two treatment decisions followed by a dataset for treatment decision timepoint 3 (T3). Treatment decision timepoints 4–6 (T4–6) consisted of one transcript per decision timepoint therefore these transcripts were analysed together.

Research studies can make comparisons between themes identified at each point to demonstrate changes over time, which is considered a concept of longitudinal research (Bennett, Kajamaa, and Johnston 2020). Longitudinal research can include analysing data from multiple timepoints to understand how a phenomenon may change over time allowing for interpretation and comparisons to be made between the different timepoints. In this study, comparisons were made between codes and themes identified at each decision point to explore how parent decision-making changed over time. This required working with multiple datasets with the need for a consistent systematic approach within each dataset (each decision timepoint) and across the whole dataset (all decision timepoints) to avoid an underdeveloped or narrow analysis. The iterative back and forth nature within and across the datasets resulted in a complex process. I was searching and analysing the concepts of decision-making which changed over time for parents whilst also considering concepts which were unique to each dataset or were present throughout decision-making irrespective of how many decisions parents had made.

Practically to manage the complexity of working within and across multiple datasets, I arranged my research time so that I had consecutive days to develop the analysis working on one treatment decision timepoint at a time. I analysed data from one treatment decision timepoint and then move iteratively back and forth across previous datasets to support inductive and deductive coding. Working consecutively without having long periods of time away from the data meant I did not lose my thought processes when combining inductive and deductive analytical approaches which I found difficult as I had two analytical processes to work with. Having this time meant I was in the moment with the data without distractions. The further into the analysis of each dataset, the more time was required for this iterative working as I had more transcripts to review moving back and forth across the transcripts from all the previous treatment decision timepoints. The time emersed in this analytical development was lengthy and took more than triple the amount of time than the analysis for T1.

From T2 onwards, data were analysed inductively and deductively to build the concept of how decision-making changed over time. Research question two was incorporated into these datasets, 'investigate whether parent decision-making changes over time and if so and how?'. Deductive coding was informed by the codes and themes from previous treatment decisions. This was an interpretative lens through which to start coding and meaning-making of the data from T2 onwards. Deductive approaches can take a theory-driven approach informed by theoretical frameworks, models or published literature relating to decision-making. Using these approaches to inform the deductive part of the analysis could have influenced the analysis of the whole dataset when moving between inductive and deductive approaches. Therefore, my deductive lens was informed by the data from within this study to avoid these influences. This interpretative lens helped me engage with the data at each treatment decision timepoint, taking the codes from previous treatment decisions and reviewing these alongside coding the current transcripts. This supported building a narrative of how parents made repeated treatment decisions to investigate whether decision-making changed over time and if so how. For example, T2 deductive coding was informed by themes and codes (Figure 10, version 5.0) from analysis of the T1 dataset. Themes and codes from T1 and T2 informed deductive analysis for T3 and so forth. This approach enabled me to see what was consistent in decision-making for parents and what was present only at specific timepoints.

Once I had deductively analysed the dataset, I started inductively coding each transcript within that dataset to analyse what was new, different, and how for parents as they made repeated treatment decisions. My inductive coding was to the entire transcript including text segments which may have been assigned to a deductive code. The purpose of this was to collate codes related to decision-making which were not specific to the number of treatment decisions parents had made. For example, being a source of support and guidance to other parents was not necessarily related to the number of decisions parents made but was seen across different treatment decision timepoints. This support and guidance related to parents sharing their experiences of a particular treatment wherever they were in their decision-making journey. The inductive codes were then reviewed against transcripts from previous treatment decision timepoints to see whether codes could be refined, revised, or regrouped as I was conscious that I may have used a different language to interpret the same meaning. To differentiate between deductive and inductive codes, the letter 'D' was added in front of the code to indicate a deductive code. In parallel with coding, I wrote notes in my reflexive journal (Box 8)on inductive codes and what appeared to be different for parents who had made more than one treatment decision.

Box 8. Reflexive diary on sense-making for inductive coding at T2.

"The cognitive adjustment differs for parents as they recognise that treatment is not working, which links with uncertainty and hope for their child's overall outcome. As children are exposed to more treatments the treatment options become limited due to the child's clinical condition and having exhausted treatment options. Complexity of this decision-making requires more time with consultants to discuss and rediscuss treatment options, one clinic appointment is not enough. Parents are more willing to consider and try early phase clinical trials as the treatment options decrease whereas previously these were not considered an option.

There are limitations of what parents are prepared to consider for treatment however as subsequent decisions are made these limitations are pushed and treatment options which were ruled out previously are rationalised and administered due to the limited options which remain. This could link with cognitive transitioning and hope."

Parent involvement and what they require as they make repeated treatment decisions differs to that of parents who have made one treatment decision and the literature to date surrounding this. The nuances increase in parallel with parents becoming conflicted in their decision-making balancing treatment with their child's quality of life.

Once I completed coding of each decision timepoint, I created a table and collated codes and associated data mirroring my approach with data from T1. The purpose of these tables was to help identify codes which had similar meaning, refining codes and grouping them for shared meaning. In the T2 analysis, I observed similarities between codes across the transcripts. For example, hope seemed prominent and I interpretated this as the need for hope as previous treatments had failed. The analysis of each decision timepoint was discussed with the two parents from the PPI group to support identification of codes which had shared meanings and could be regrouped. Parents acknowledged the potential for recall bias as parents made repeated treatment decisions. For example, parents were more likely to provide an accurate account of their latest treatment decision, whereas with previous treatment decisions the treatment outcomes were known which may have influenced their perceptions of decisions made at the time. This supports a relativistic and cognitive constructivism approach which is time and content dependent, influenced by parents' cognitive and emotional states given that previous treatment has not worked, with the need to make further treatment decisions.

Treatment decision timepoint two

Deductive codes at T2 were indicated by an asterisk and highlighted in yellow on the thematic map from T1 (Figure 11) to indicate which were consistent at T2. An inductive thematic map (Figure 12) was devised with one additional theme, 'parent experience increases as more treatment decisions are made', and the theme 'cognitive adjustment as treatment strategies fail' was an extension of the cognitive adjustment theme seen at T1. For a robust approach, new inductive codes at each timepoint were explored in data from previous treatment decision timepoints. This approach strengthened analysis, reviewing these codes to see whether they were present in previous data but missed during analysis or supported changes in decision-making over time.

Treatment decision timepoint three

Deductive codes at T2 were highlighted in pink on the thematic map from T1 and T2 (Figure 13 and Figure 14). There were 12 new inductive codes from T3 analysis (Figure 15). However, some of these codes appeared to be an extension of existing codes and themes. For example, the code 'parent knowledge and understanding of disease and treatment increases over time, which informs their decision-making' extended from the code 'decisions informed by knowledge, experience and research' seen at T1. My analytical interpretations whilst analysing the T3 data showed

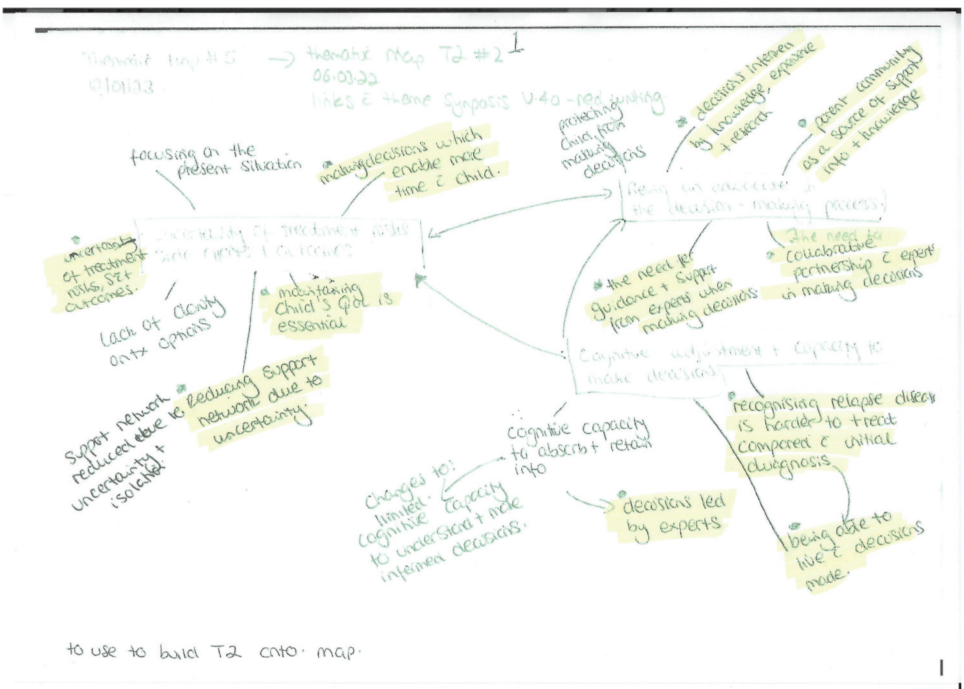

Figure 11. Thematic map deductive codes T2 on T1.

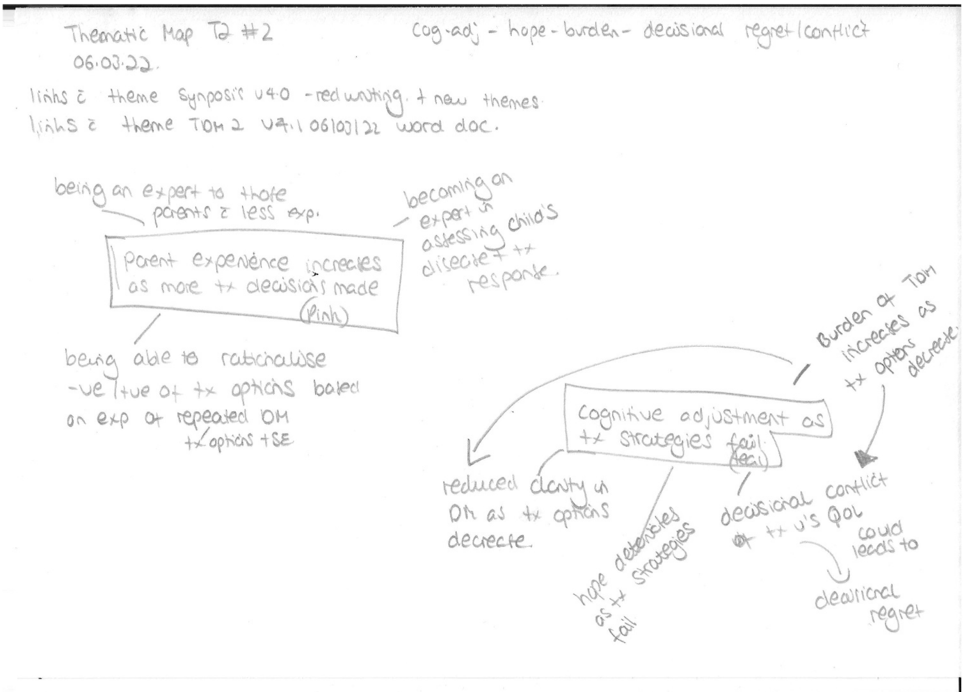

Figure 12. Thematic map for T2.

there was a 'novice to expert' component in decision-making where parent involvement and opinions on treatments developed over time. Parents gained more experience and knowledge as they made repeated decisions informed by their research on treatment options and lived experiences. This knowledge and experience were used to rationalise their approaches to decision-making, continuing with treatment to have more time with their child and potentially cure them. At this stage, the 12 inductive codes were not developed further as some were felt to be extensions to pre-existing codes and themes seen in T1 and T2. This required consideration as to how to present this within the narrative and to explore whether the remaining data supported or refuted these concepts.

Treatment decision timepoint four to six

Deductive codes at T4–6 were highlighted in green on the thematic map from T1, T2 and inductive codes from T3 (Figures 16, 17 and 18). There were 11 new inductive codes from T4–6 (Figure 19). I wrote a synopsis of inductive codes at T3 and T4–6 to support my analytical thinking. This led me to see codes were linked to the three themes from T1. For deductive data, I wrote a synopsis which added to pre-existing themes at T1 and T2. This explored my thinking whether data is related to the concept of decision-making over time

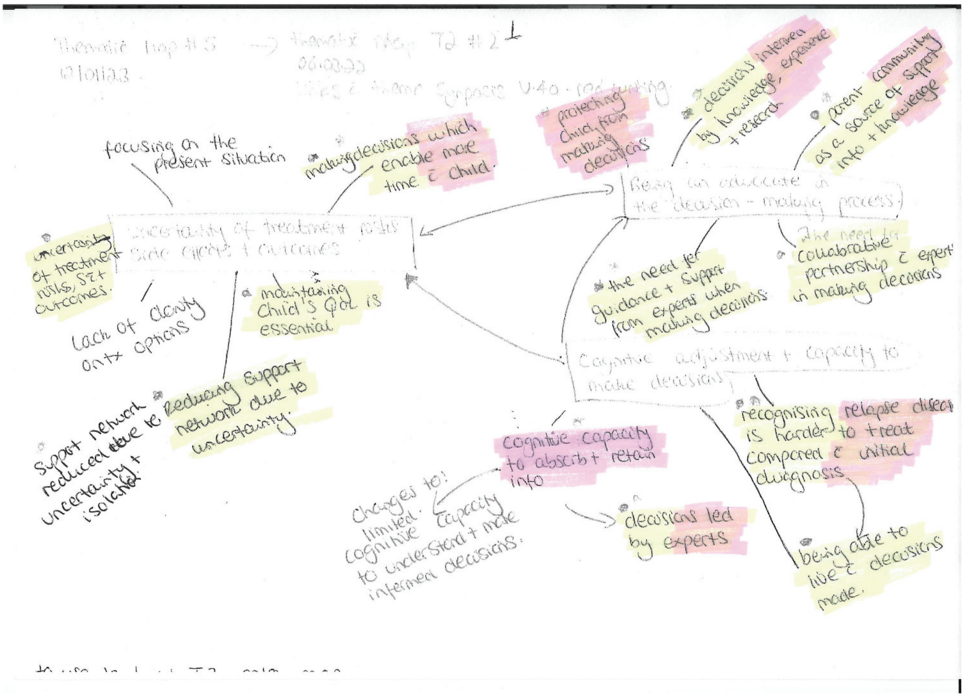

Figure 13. Thematic map deductive codes T3 on T1.

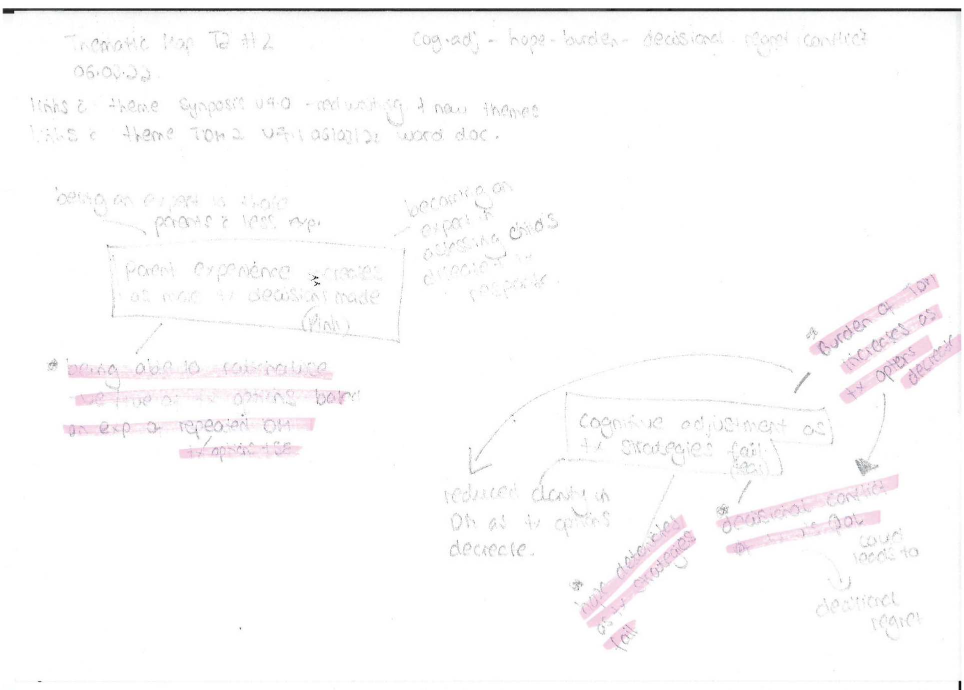

Figure 14. Thematic map deductive codes T3 on T2.

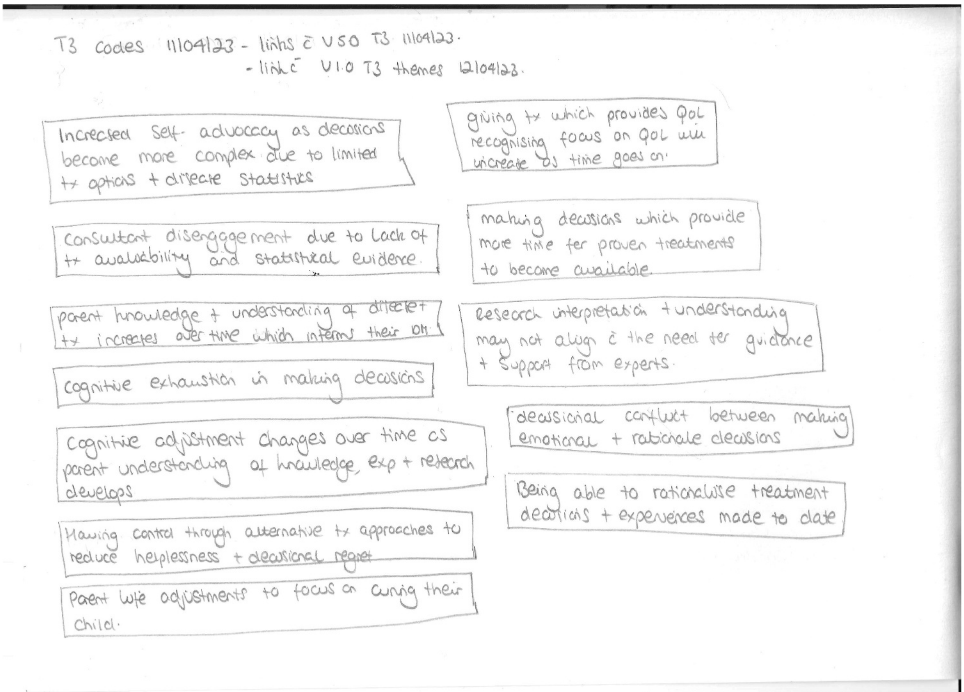

Figure 15. Inductive codes from T3.

or decision-making in general, not specific to making repeated treatment decisions. This helped with my analytical thinking of concepts which changed over time, for example, uncertainty and experience. I drew a diagram of the novice to expert continuum in parent treatment decision-making and codes which informed how this decision-making changed over time (Figure 20). This visual helped clarify my understanding of different concepts related to treatment decision-making for parents. I discussed the diagram with my supervisors who suggested drawing out the themes and their connections with each other (Figure 21). This diagram was used to support interpretation of my findings and write-up of the analysis in conjunction with theme synopses written throughout my analytical journey. I also read literature and theory related to uncertainty, time, knowledge, and decision-making to situate these findings in relation to wider literature and theory.

Phase six: writing up

This phase is the detailed shaping of the narrative acknowledging that even at this stage of the analytical process, the researcher can still be refining their analysis and making changes as they write (Braun and Clarke 2022b). My narrative account of the analysis included data extracts related to the research questions and an interpretation of the analysis contextualising this with

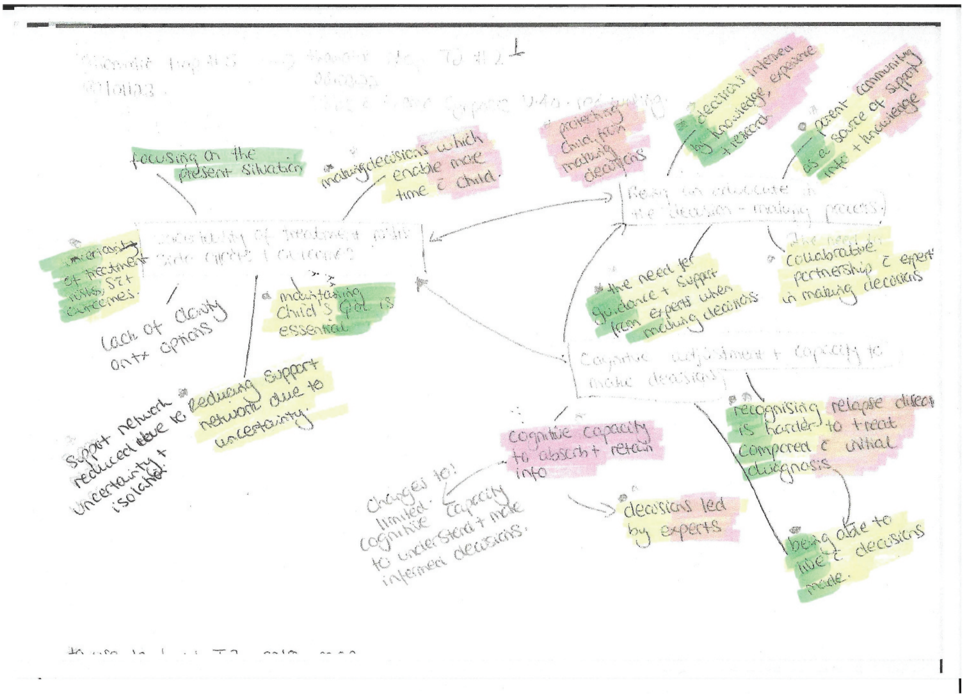

Figure 16. Thematic map deductive codes T4–6 on T1.

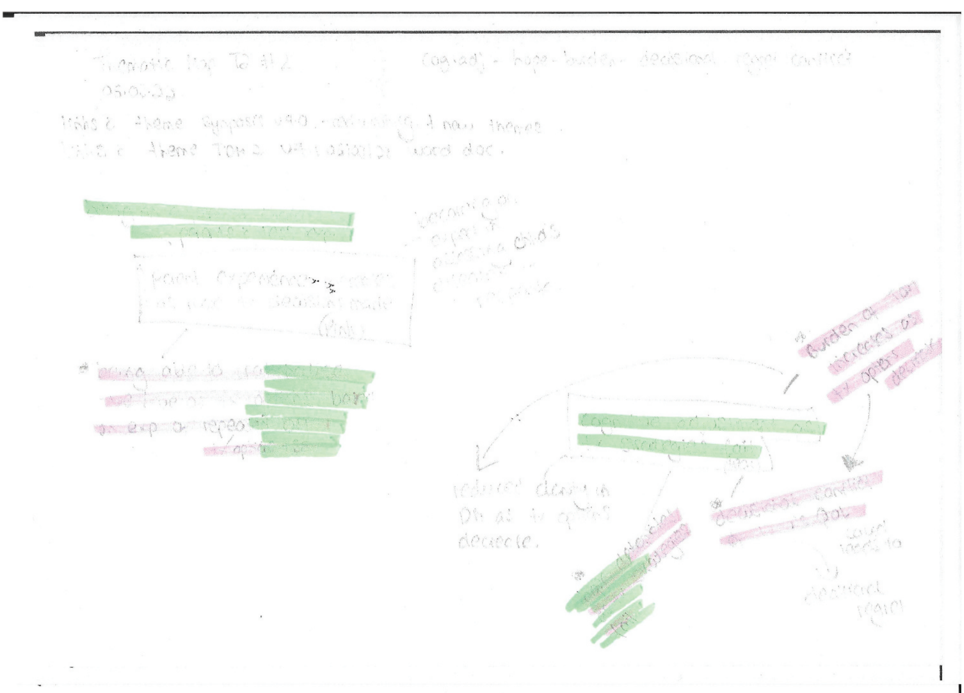

Figure 17. Thematic map deductive codes T4–6 on T2.

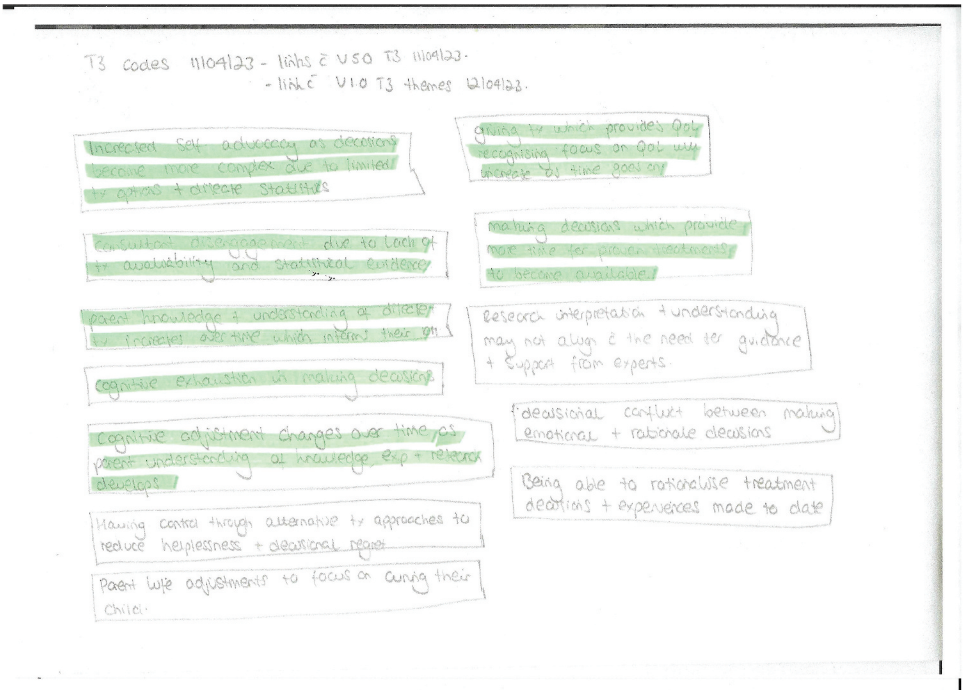

Figure 18. Deductive codes T4–6 on inductive T3 codes.

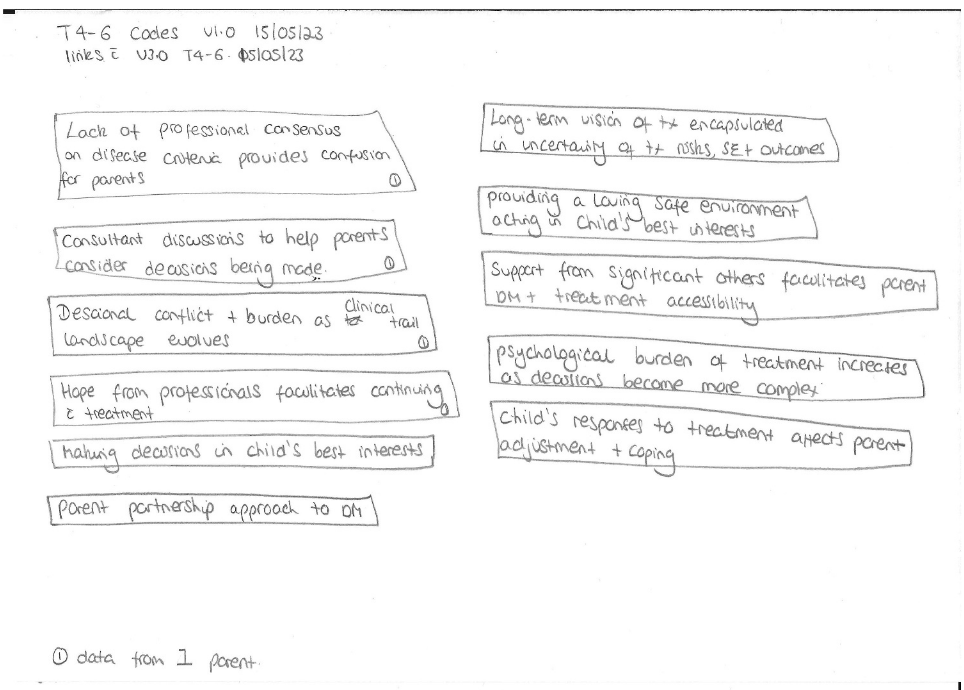

Figure 19. Inductive codes from T4–6.

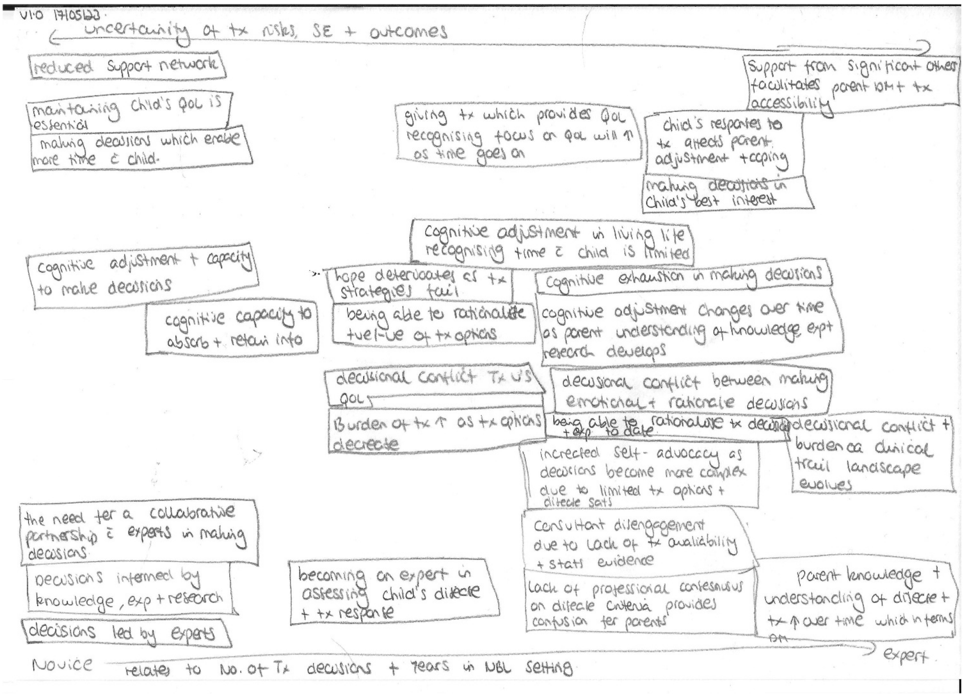

Figure 20. Novice to expert continuum of decision-making.

known research and theory. I developed a conceptual framework which showed the relationships between themes and how these themes influenced, informed and enabled parents who were making repeated treatment decisions when their child had relapsed or refractory neuroblastoma. This work required numerous iterations to write up a cohesive narrative for publication which took considerable time and cognitive thinking. The publication structure and content were guided by the Braun and Clarke (2020a) tools for evaluating thematic analysis manuscripts for publication.

Theme synopsis supported the write-up stage, which started during phase three of the analysis for parents who had made one treatment decision. Theme synopses provided a written narrative account of the story on what was happening within each theme and how themes were related. I expanded the narrative developed at each treatment decision timepoint of the analysis to address the research questions and cohesively bought this together to describe the concepts of decision-making which changed over time for parents. Once I had shaped the narrative of the analytical write up, I worked on the data extracts from participants which supported the narrative following Braun and Clarke's selection tips (2022b). For example, I ensured I included extracts from a range of participants and edited unnecessary details. The findings from this study have been published (Pearson et al. 2024)

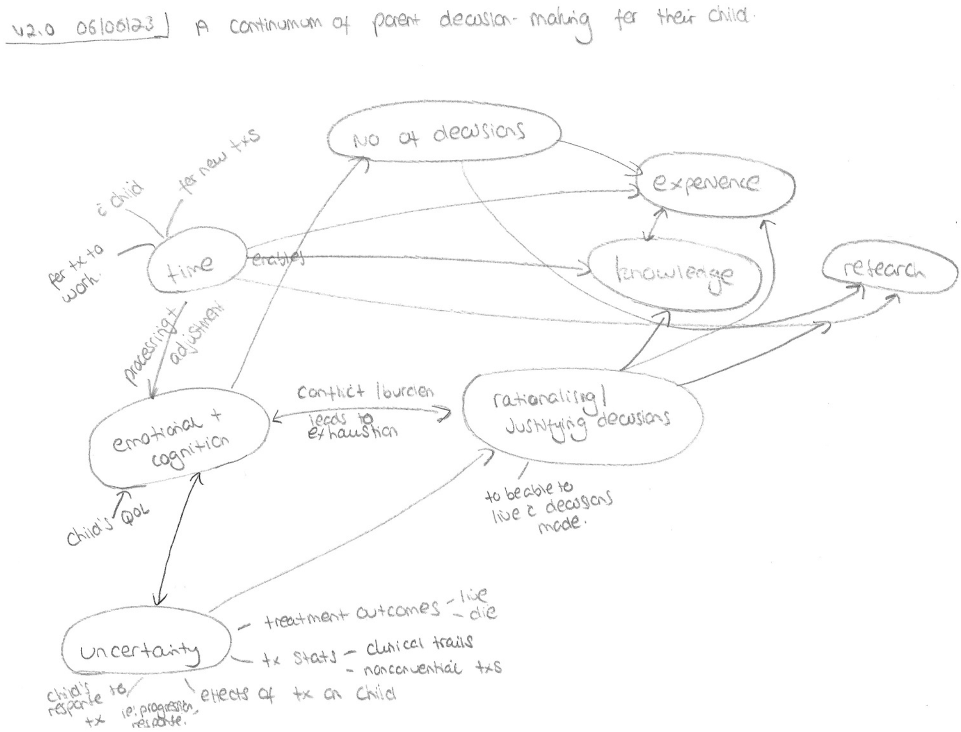

Figure 21. Continuum of parent treatment decision-making.

Conclusions

This paper makes a novel contribution in encouraging researchers to be transparent in the analytical process of qualitative data analysis by providing a worked example of the analytical process taken by a novice qualitative researcher. The purpose of sharing this example is to support researchers understanding of the process and approach to reflexive TA in complex data sets. This work highlights the complex nature of qualitative data analysis of multiple datasets combining inductive and deductive approaches, which was time-consuming and required researcher patience. Although there are software programmes which support the organisation and management of qualitative data, the researcher's positionality, subjectivity and reflexivity and their investment in the analysis process is crucial to the reflexive TA method. This differs to quantitative research where there are software programmes which can assist with developing the analysis for you.

Reflexive thematic analysis is 'methodologically flexible', open to interpretation and acknowledges the researchers' subjectivity in the analysis. While it can be a complex method for a novice to grasp, the array of Braun and Clarke webinars, publications, and most recent textbook (Braun and Clarke 2022b) provided a solid understanding of the reflexive TA approach and ways to work with this in my own analysis. The importance of re-reading and reviewing these materials whilst undertaking the analysis helped clarify my thoughts and approach which at times made me see I was ready to move onto the next step of analysis or further iterative work was required.

Despite reading all these materials and acknowledging the iterative and sometimes messy nature of reflexive TA, I had not fully appreciated the impact this would have on me as someone who is uncomfortable with 'messiness' and likes things tidy! The iterative back and forth process between phases 2–5 with refining codes, reviewing, and defining themes was time-consuming, complicated, and confusing at times. The importance of documenting my approach and interpretations of data in my reflexive journal helped my critical thinking and ensured I was systematic and robust in my approach. This was essential with the large volume of data to analyse with different thoughts and interpretations as it could have been easy to forget and lose focus. Talking through my analysis with parents from the PPI group and my supervisors was helpful in critically scrutinising my interpretations and narrative of the data. Each time, this provided me with further insight to refine and review my analysis, which was pivotal to providing a cohesive narrative. Based on my experiences as a novice qualitative researcher using reflexive TA in analysing multiple datasets, I have developed the following recommendations:

- Keep a notebook of your analytical process and reflexive accounts.
- Be prepared to be uncomfortable and unsure of where your analysis is taking you . . . . This is all part of the journey!
- Be fluid in your approach so that you can move iteratively back and forth between the phases to develop your analysis to the best of your abilities.
- Spend time away from the data as this provides a new perspective to take your analysis forward.
- The time you allocate to your analysis is never long enough – double this at least!
- Talk through your analysis with others, this helps clarity in your analytical thinking.
- Be clear on how you are approaching the deductive part of your analysis – what is this driven by?
- Once chosen, what is driving your deductive approach ensures that the interpretative lens you use is consistent across multiple datasets of deductive analysis.
- Working with multiple datasets is mentally consuming, a robust audit process to support iterative back and forth across and within datasets is required.

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Data availability statement

Data is included which supports the approach and process of reflexive thematic analysis as a worked example. Data from the interviews can be obtained from the main author. The empirical findings from the study which this example uses will be published in a peer-reviewed journal.

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Appendix

Table 4: **Extract of code and theme development for treatment decision timepoint 1**

Research Question:

Identify, describe, explain and explore how parents make treatment decisions when their child has relapse or refractory neuroblastoma (RQ1)

Codes	Grouping of codes	What does this mean? Theme Development	Participant Quotes	Literature to Support/ Refute
Child's response to diagnosis based on previous experience Child's limited involvement in TDM Child's understanding of the disease Filtering information to child on the reality of the situation	Protecting child from the situation	Protecting child from the burden of making decisions, Child's age as a barrier to cognitive understanding in decision-making *?links with advocating*	"This time round, he was really devastated to get the diagnosis again because he knows what's coming in terms of treatment . . . . He had quite a few more questions this time, but I think he's still young enough that he's not really involved in making the decisions. He was more involved in the discussion with the consultant this time and the consultant took the time to explain to him why he was deciding on this treatment what it was for, but ultimately it was still out decision to make. It wasn't really something that we gave ***** an option on" "He likes to be in charge. We do joke, he thinks the world revolves around him and things. So I think that would be a mistake, to give him the impression that he has a say in whether we do treatment or we don't do treatment. I think that's not for him to decide or have that responsibility" "He knows he's unwell. I have told him he's got a new tumour, so we call it the monster. Apart from that, no, and it's hard not to tell him because I told him everything from your going down for a scan, your going to have an injection today . . . . I feel like that's how he's become so brave, but obviously now, to look at him I feel really guilty. You can't tell your child, but this is information I have to hold back"	Literature on being a good parent – Hinds et al

(*Continued*)

(Continued).

Codes	Grouping of codes	What does this mean? Theme Development	Participant Quotes	Literature to Support/ Refute
Giving treatment which is feasible to living life Child's potential limitation to tolerating treatment QoL more important than treatment Balancing QoL with Tx intensity Child wanting to be well Treatment given is dependent on child's wellness	Child's QoL central to choosing a treatment *Links with*: *protecting child from the situation*	Advocating for child	"What takes centre stage is what's the best treatment for the child who needs the treatment. Then it's got to also fit around other things. Things have got to be feasible aswell" "So for us whichever works the best, and at the same time if our son can handle it. I don't want to reach the situation where this is the best treatment yes, but my son cannot handle it" "I think for us, not putting ***** through really, really intense treatment from the get-go. I mean obviously is she needs it then we will obviously make decisions around that, but for the time being if something is working and it's providing her with that quality of life, we're more than happy to continue" "We've got in the back of our minds, that if it came to the point where we like, this isn't worth it, then that would change, but that hasn't happened yet. It's just been whatever is the most likely to work, we go with that really" "I'm quite happy to do anything as long as she's well. If it means that I'm in hospital with her the whole time, I'll be there" "We'd done our research, a lot of research actually. Yeah, our decision was literally, we'll try anything as long as ***** well with it, sort of thing" "He wanted to be able to go back to school, which he's done in the last few weeks a little bit, just as much as he's been able to"	